



CrossLinkU-NX SoM for HMI Demonstration

Reference Design

FPGA-RD-02333-1.2

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Abbreviations in This Document

A list of abbreviations used in this document.

Abbreviations	Definition
AI	Artificial Intelligence
CM5	Compute Module 5
CNN	Convolutional Neural Network
CPU	Central Processing Unit
Demo Design	Demonstration Design
ECDSA	Elliptic Curve Digital Signature Algorithm
EVE	Edge Vision Engine
FFC	Flat Flexible Cable
FM	Firmware Management
FOV	Field of View
FW	Firmware
GPIO	General Purpose Input/Output
HMI	Human Machine Interface
IPS	AI Inference Per Second
JTAG	Joint Test Action Group
micro-ISP	Micro Image Signal Processing
OPN	Ordering Part Number
OS	Operating System
PC	Personal Computer
RPi	Raspberry Pi
SDK	Software Development Kit
SoM	System on Module
UI	User Interface
ULP	Ultra-Low Power
WiFi	Wireless Fidelity

1. Introduction

This user guide introduces the CrossLinkU™-NX System-on-Module (SoM) HMI Demonstration Design, purpose-built for Human-Machine Interaction (HMI) applications. The design enables AI-driven interaction capabilities, including person detection and face recognition enabling intuitive and intelligent user experiences. Optimized for edge device processing, the solution supports ultra-low power mode, while adhering to industry standards for embedded vision and AI systems.

If you plan to use the CrossLinkU™-NX SOM to enable HMI-based interactions in a custom application, refer to the [Communication Protocols and the EVE API Guide for the CrossLinkU-NX SoM \(FPGA-UG-02250\)](#) document, which describes several methods for reading the metadata, allowing you to choose the approach that best meets your objectives and requirements.

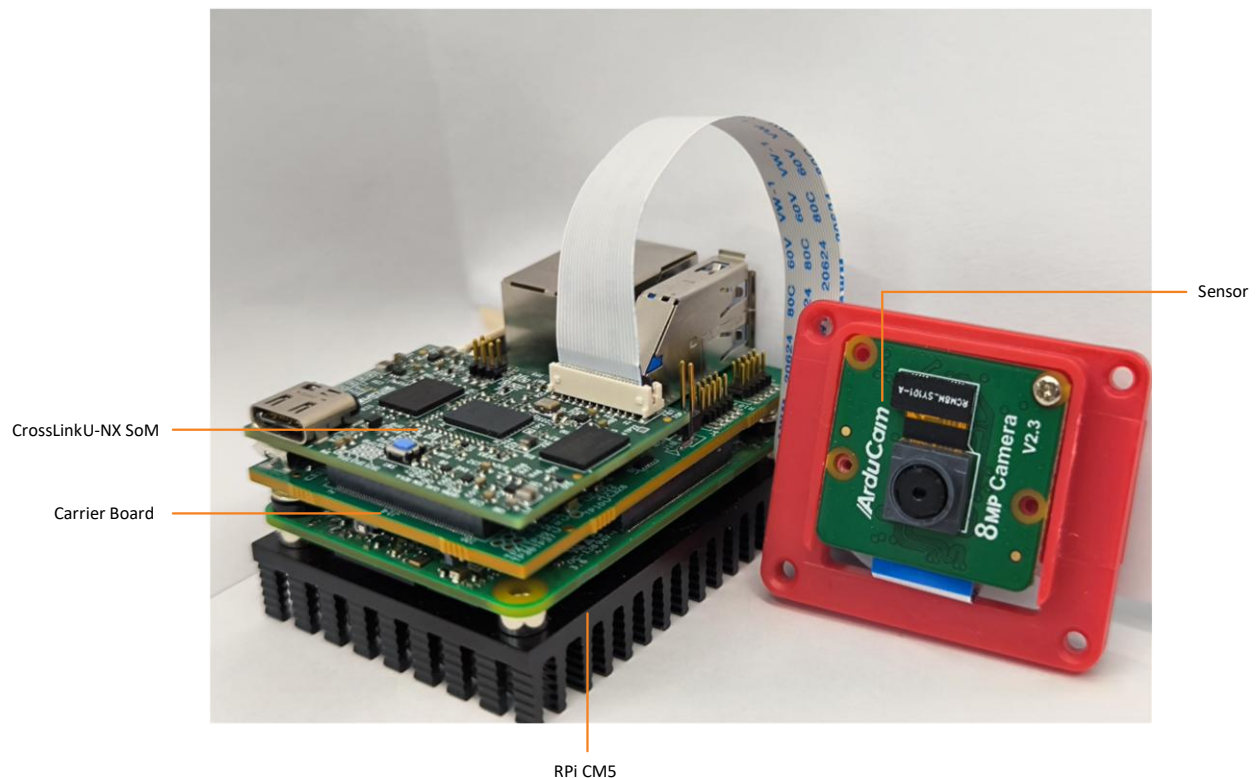


Figure 1.1. CrossLinkU-NX SoM Hardware Components for HMI Demo

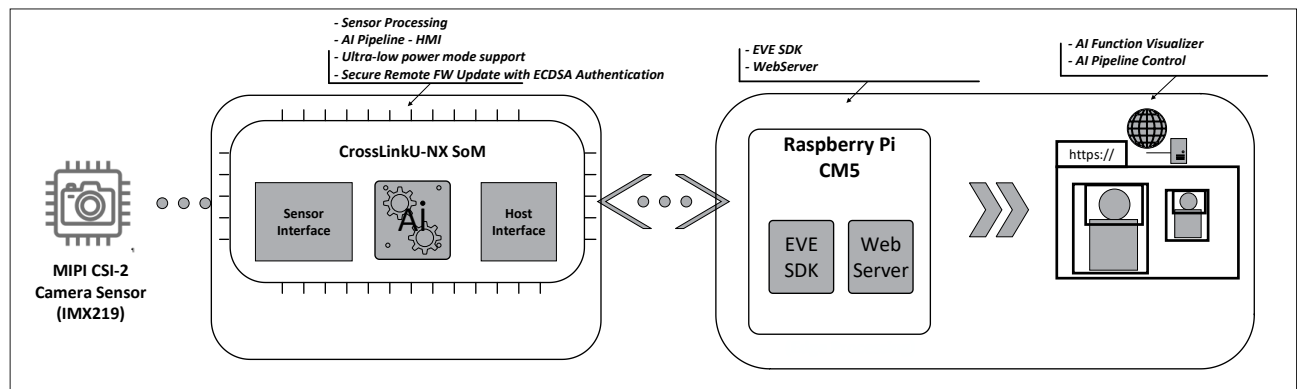


Figure 1.2. System Overview of CrossLinkU-NX SoM for HMI Demo

The key components of the CrossLinkU-NX SoM HMI Solution are:

- CrossLinkU-NX System on Module HW board (LIFCL-33U-SOM-EVN)
 - Powered by the Lattice CrossLinkU-NX with HMI Demo Firmware.
 - Performs sensor MIPI streaming to host system
 - Integrated AI pipeline with CNN Plus IP for HMI detection
 - Implements Low-power AI processing leverage CrossLinkU-NX AON controller
- Generic Carrier HW Board (LF-GEN-CR-PCBA)
 - Interfacing between CrossLinkU-NX SoM HW Board and Raspberry Pi (Rpi) CM5
 - On-board FTDI Controller for:
 - PortA: CrossLinkU-NX SoM and CrossLinkU-NX JTAG for Configuration
 - PortB: Reserved
 - PortC: Reserved
 - PortD: CM5's UART Serial Console Access (Setup)
 - Bridges CM5 USB-C port, USB-Type A port and Ethernet Port
- Raspberry Pi Compute Module 5
 - Equipped with USB, Ethernet, and GPIO interfaces
 - Acts as a host to handle AI metadata processing with EVE SDK and visualization dashboard with the web server
- Sensor
 - Targeting MIPI CSI-2 camera sensor IMX219
 - ArduCam B0191, Narrow FOV, Coverage 62.2°(H) × 48.8°(V)

1.1. Quick Facts

Download the HMI demonstration files from the [HMI Demonstration](#) web page.

Table 1.1. Summary of the Demonstration Files

General	Target Devices	CrossLinkU-NX (LIFCL-33U-FCCSP104)
	Source format	Bitstream
Software Requirements	Software tool and version	• Lattice Radiant™ Programmer 2025.1 or later • Raspberry Pi Imager v1.9.6 • Raspberry rpiboot
	IP Version (if applicable)	Not applicable
	Host Software	• Raspberry Pi OS – Bookworm¹ • Lattice EVE SDK • Lattice Web Server • PuTTY or equivalent (for example, Tera Term)
Hardware Requirements	Demo Kit	CrossLinkU-NX SoM Demo Kit (LIFCL-33U-SOM-EVN)
	Host	Raspberry Pi CM5

Note:

1. Lattice Custom RPi OS for CrossLinkU-NX SoM HMI Demo. OS update is disabled by default.

1.1.1. Quick Start Guide

Refer to the [Running the HMI Demo Design on CrossLinkU-NX SoM HW Board](#) section for instructions on setting up and running HMI demonstration on the Raspberry Pi CM5 with CrossLinkU-NX SoM.

1.2. Features

The CrossLinkU-NX SoM Demo Design – HMI Demonstration offers the following key capabilities:

- Sensor Interface
 - Support 2-lanes MIPI CSI-2 Camera Sensor (IMX219), Narrow FOV
 - Support MIPI Streaming to Host for Camera Preview
 - Support MIPI for AI Processing
- AI Pipeline
 - Integrated AI pipeline utilizing CNN Plus IP for edge AI inference
 - Support AI Features such as Person Detection, Face Detection, Face Landmark Detection, and Face Identification
 - Metadata output through I2C with interrupt event signaling
 - Optimized for Low-power architecture with CrossLinkU-NX AON support
- Host
 - Runs on Raspberry Pi OS – Bookworm
 - Enables real-time AI visualization using the Lattice EVE SDK, with web server interface support
 - Adds plug-and-play Ethernet connectivity with an out-of-the-box DHCP server on the host
 - Provides real-time visualization of sensor telemetry
- Firmware Update
 - Support remote firmware update through Lattice Firmware Management Application
 - Secure firmware update mechanism with ECDSA authentication

2. Directory Structure and Files

Table 2.1 illustrates the folder description of the CrossLinkU-NX SoM HMI Demo Package. Comprehensive information is available in the GitHub README file, which is linked on the [HMI Demonstration](#) web page. The file names may vary across different packages. Refer to the *GitHub README* file for the download instructions.

Table 2.1. Folder Description

Folder/File	Description
CLNX_SPI_IMAGE ¹	This folder contains the 64 MB CrossLinkU-NX SoM SPI Flash Binary.
EVE_WEBSERVER ¹	This folder contains the installer for EVE and Webserver.
FW_UPDATE_TOOL ¹	This folder contains the CrossLinkU-NX SoM Firmware Management application.
RESOURCES ¹	This folder contains the resource files and scripts.
RESOURCES/RPI_SCRIPTS ¹	This folder contains the scripts used for the demo and programmer scripts.
RPI_DRIVERS ¹	This folder contains the Lattice drivers and overlays used for the demo.
RPI_OS	This folder contains the Raspberry Pi OS image. ²

Notes:

1. These contents are preinstalled into the Raspberry Pi OS Image.
2. Lattice Custom RPi OS for CrossLinkU-NX SoM HMI Demo. OS update is disabled by default.

3. Functional Description

This section describes the architecture of the HMI demonstration with Lattice CrossLinkU-NX SoM. The system architecture of the demonstration design involves several key subsystems working together seamlessly. There are five main subsystems involved in the design:

- MIPI Sensor Processing
- AI Detection
- EVE SDK
- Web Server
- FW Update

Figure 3.1 shows the overview of the HMI demonstration design block diagram and five main subsystems.

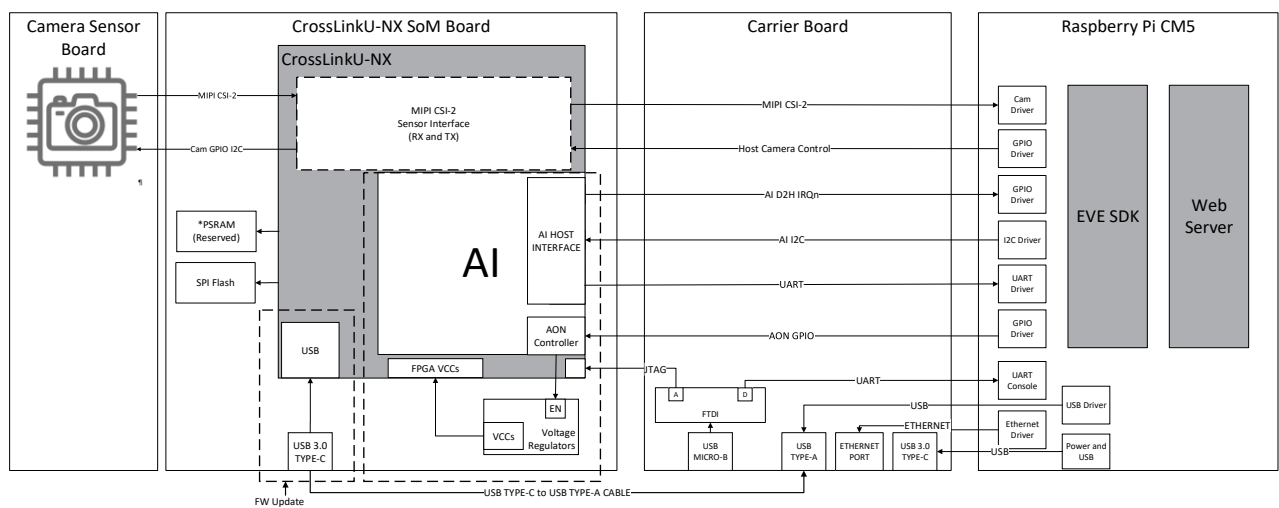


Figure 3.1. Overview of HMI Demonstration Design Block Diagram

3.1. MIPI Sensor Processing

The design implements 2-lane MIPI CSI-2 sensor processing. In the *Streaming* mode, the design supports MIPI passthrough, enabling uninterrupted video transmission from the camera to the host system. Concurrently, the camera stream from MIPI RX is fed to AI processing pipeline for HMI AI Inference.

Table 3.1. MIPI Sensor Processing Features

Feature	Specification
Camera Sensor Module	Narrow FOV: IMX219 – ArduCam B0191
MIPI	RX and TX: CSI-2, 2-Lanes, RAW10, up to 960 Mbps per lane.

3.2. AI Detection Module

The AI pipeline leverages CNN Plus IP to deliver accelerated inference for edge AI applications. The supported models include Person Detection, Face Detection, Face Validation, and Face Identification. The pipeline incorporates pre-processing (micro-ISP), Machine Learning execution, and post-processing stages. AI inference results are transmitted as metadata over AI I2C, accompanied with interrupt signal to Host for event notification. Metadata includes bounding box coordinates, distance from the camera, head pose orientation and others. The AI pipeline supports ULP (Ultra-Low Power) mode which utilizes the CrossLinkU-NX AON Controller to further optimize the power consumption.

Table 3.2. Supported ML Models

Model	Function
Person Detection	Detects human bodies
Face Detection	Detects the human face
Face Validation	Detects and validates face landmarks
Face ID	Face identification

3.2.1. AI Pipeline

The AI pipeline consists of an infinite loop of inferences and a metadata output. Every loop in the basic configuration performs the following:

- Run Person Detection
- Run Face Detection
- Run Face Validation
- Run Face ID
- Output Metadata

3.2.2. AI Configuration

The AI pipeline can be configured from the Host at runtime. For each supported model, it can configure:

- Enablement
 - By default, each AI model is enabled.
 - If a model is enabled, its output is present in the metadata sent to the host at the end of the loop.
- Maximum Inference Per Second (IPS)
 - By default, the Maximum IPS of each model is MAX (value of 0). It runs as fast as possible with no restriction.
 - If it is set to something different than 0 for a specific model, the pipeline may not run the model on every loop to prevent exceeding the Maximum IPS value.
 - The pipeline can pause at the end of a loop to not output faster than the highest Maximum IPS out of all the models.

In ULP mode, the AI model control and maximum IPS option are disabled. The pipeline assumes full control of the IPS. This mode is ideal for showcasing energy-efficient operation.

3.3. EVE SDK

The EVE SDK is a software library that parses HMI AI metadata and works alongside the CrossLinkU-NX SoM to provide real-time insights and structured visualizations. It processes metadata from the AI pipeline and formats it for easy visualization and analysis. When an AI interrupt occurs, the EVE SDK signals that new data is available. This data is then parsed into a defined structure, including bounding boxes, distances, head pose orientation, and other details that overlay the current image.

3.4. Web Server

The Raspberry Pi CM5 hosts a web server that provides an interactive dashboard for monitoring and control. Through this dashboard, users can visualize AI outputs, review system performance metrics, and configure the AI pipeline at runtime to meet application requirements. It also displays live readings of energy sensor data.

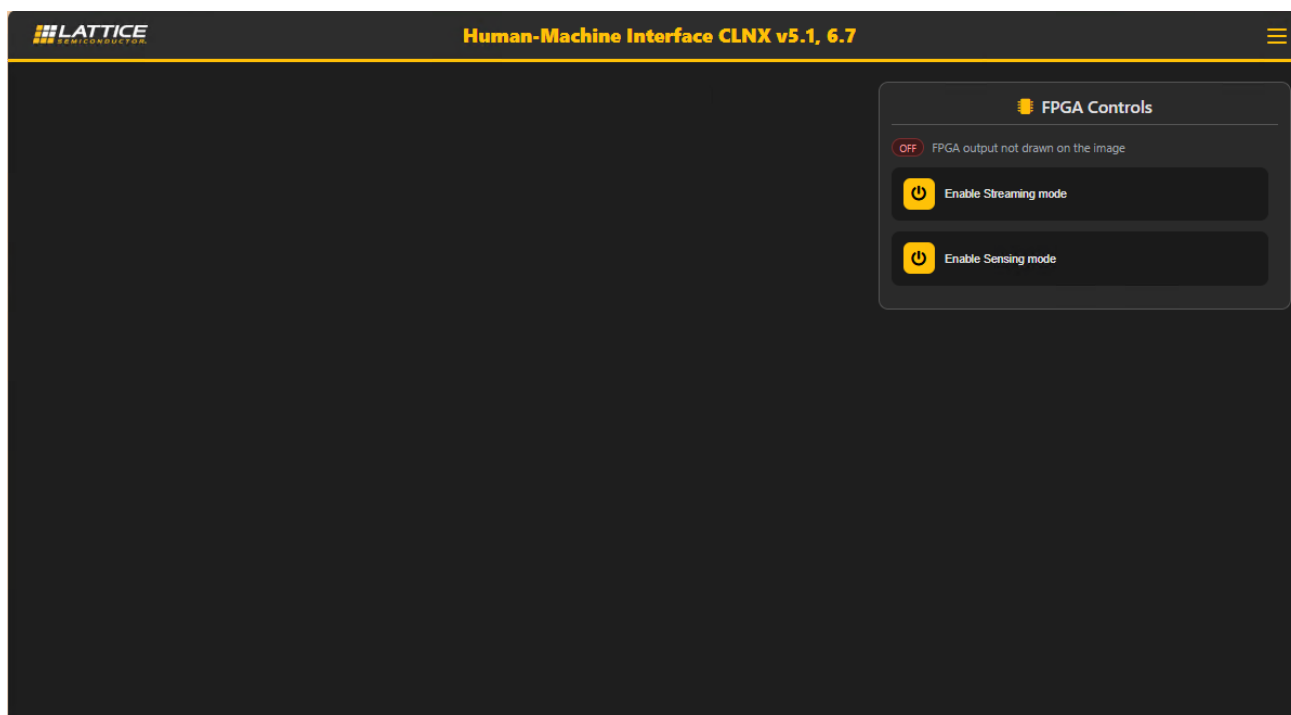


Figure 3.2. Web Server Landing Page

The HMI demo design supports three operational modes (Streaming, Sensing, and ULP Sensing) as described in [Table 3.3](#). These modes can be configurable through EVE SDK and Web Server.

Table 3.3. Operational Modes

Mode	Description
Streaming	Continuous MIPI camera streaming to host while concurrently running AI models.
Sensing	AI detection only, without video passthrough.
ULP Sensing	Ultra-Low Power Always On at ~15mW ¹

Note:

1. During ULP Standby Operation.

3.5. Remote Firmware Update

The HMI demo design supports secure and authenticated firmware updates through USB, ensuring system integrity and resilience against unauthorized tampering. When a firmware update initiated, the system transitions into Firmware Management mode, receives new firmware from USB interface to program into SPI flash and authenticates them using ECDSA signatures. Once the process is completed, the system jumps to verified binaries. During the update process:

- Camera streaming and AI operations are temporarily paused.
- These functions resume only after the firmware update is successfully completed.

A USB cable is required to connect the CrossLinkU-NX SoM HW board USB-C port to the generic carrier HW board's USB Type-A port. This connection is essential for performing firmware updates. The update process uses the Lattice Firmware Management Application. Refer to the [Running the Firmware Management Application](#) section for further instructions.

4. sensAI Edge Vision Engine Tool

The Web Server hosts a graphical web application built on the EVE SDK. This section provides an overview of each screen in the application and explains how to navigate and use its features. The Web Server provides a web application that wraps the EVE SDK with a graphical interface. The following sections describe each screen and how to use them.

4.1. Landing Page Screen

This is the initial screen displayed when the web server starts and the address is accessed through a browser. From this screen, the *Streaming Mode* can be selected to display the camera's live feed, or the *Sensing Mode* to display the metadata only.

Notes:

1. Terminate the Web Server: After completing the operation, terminate the web server by pressing CTRL+C in the terminal where the application was launched.
2. Switching Modes: To switch between streaming and sensing modes, first terminate the web server. Then, relaunch the web server to return to the landing page.

Figure 4.1 shows the first screen that appears after launching the web server.

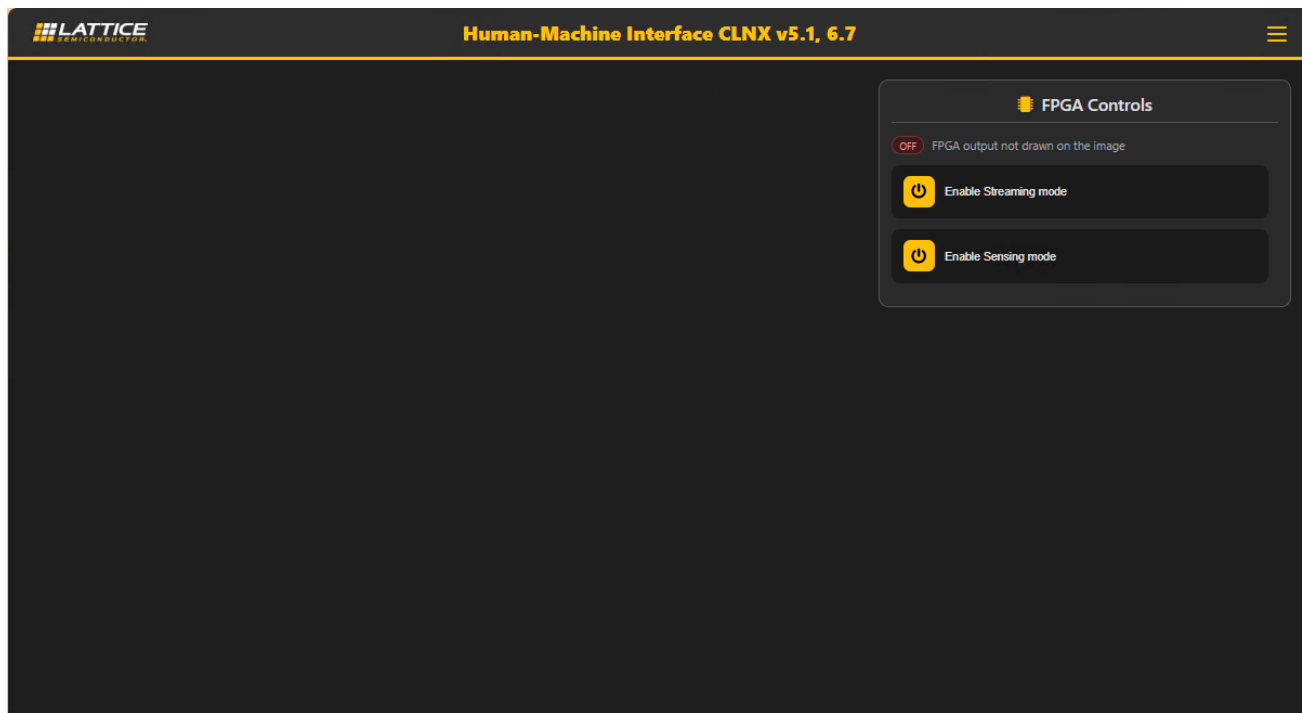


Figure 4.1. Landing Page Screen

4.2. Metadata Screen

This screen overlays metadata from the AI pipeline on the Live Video panels.

The Live Video panel updates based on the mode selected in the Initialization Screen:

- Streaming Mode – Displays live video with a metadata overlay, intended for applications that require video streaming and AI inferencing.

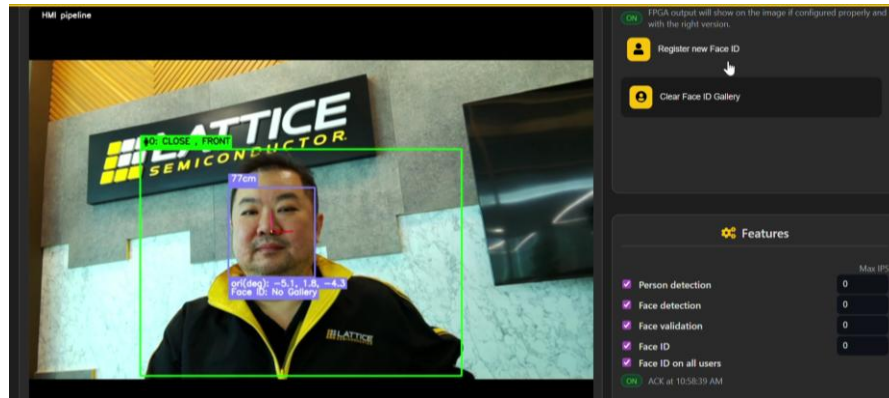


Figure 4.2. Metadata Screen – Streaming Mode

- Sensing Mode – The interface renders AI metadata over a neutral gray background. This mode is designed for applications that do not require video streaming and prioritize low-power operation. Additionally, it enhances privacy by ensuring the video feed is not transmitted to the host.
- ULP Sensing Mode – Within *sensing* mode, you can switch to Ultra-Low Power (ULP) mode using the Enable/Disable ULP Mode button. Once ULP mode is activated, all feature controls are disabled, as this mode is optimized to maintain the expected low-power operation.

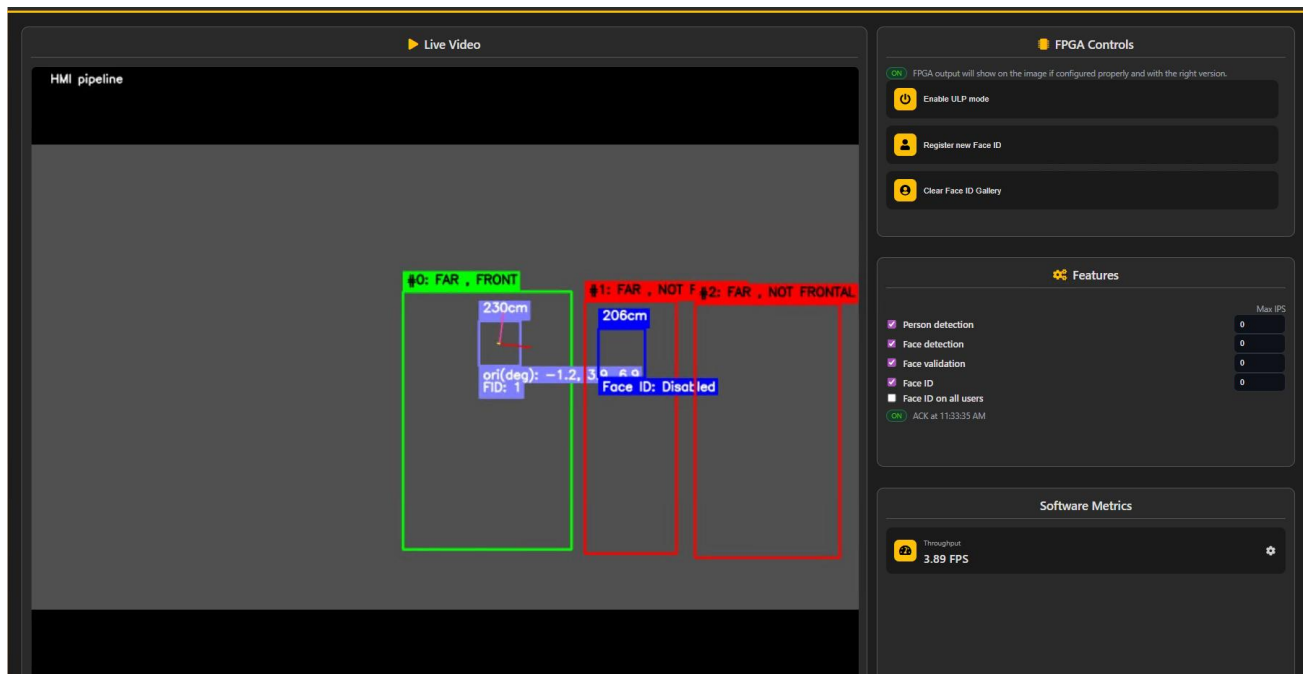


Figure 4.3. Metadata Screen – Sensing Mode/ULP Sensing Mode

4.2.1. Metadata Presentation

The metadata is displayed using colored bounding boxes combined with descriptive text. There are two types of bounding boxes used: Person Detection and Face Detection.

- Person Detection Bounding Box
 - Colour Codes:
 - *Green* for main user
 - *Red* for secondary person
 - Top Label Text: Shows *#INDEX: DEPTH, POSE*
 - INDEX: Person Index. Ranging from 0 to 4.
 - DEPTH: Relative distance of the person from the camera, categorized as *CLOSE* or *FAR*.
 - POSE: Orientation of the person's body, with possible values: *FRONT* or *NON FRONTAL*.
- Face Detection Bounding Box
 - Colour Codes:
 - *Light Blue* for the Main User.
 - *Dark Blue* for Secondary Persons.
 - Top Label Text: Display distance from Camera in cm.
 - Bottom Label Text:
 - First Row – Shows *Yaw, Pitch, Roll* in degree.
 - Second Row – Indicates Face ID status, which may include: Unregistered, FID Number, Requirement unmet, Disabled, No Gallery.

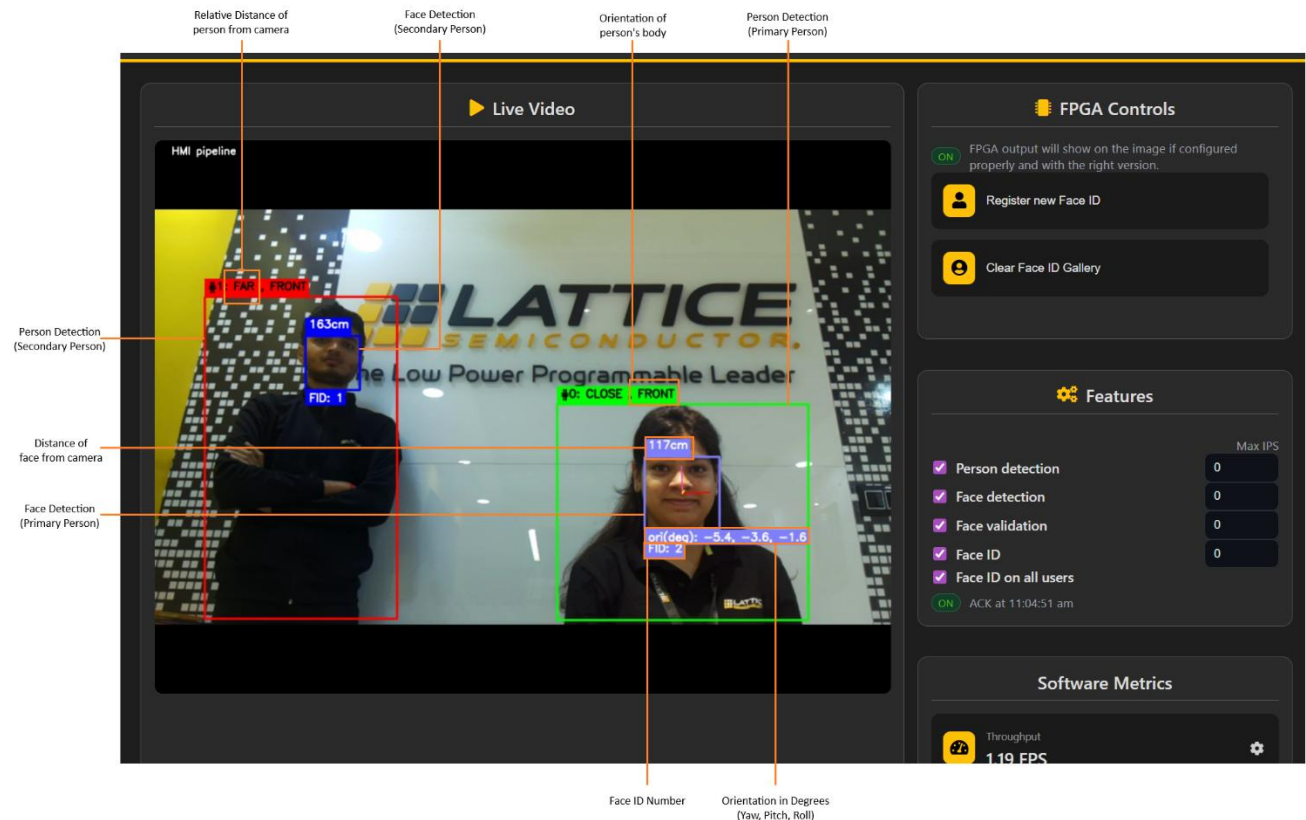


Figure 4.4. Metadata Presentation

4.2.2. Feature Enablement

The checkboxes dictate if a feature needs to run. By default, all the features are running. If a feature depends on other features to run, it silently enables those features without displaying it in the user interface.

- Face Validation depends on Face Detection
- Face ID depends on Face Detection and Face Validation

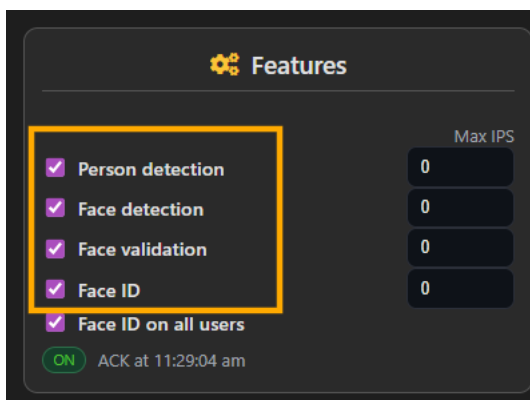


Figure 4.5. Feature Enablement

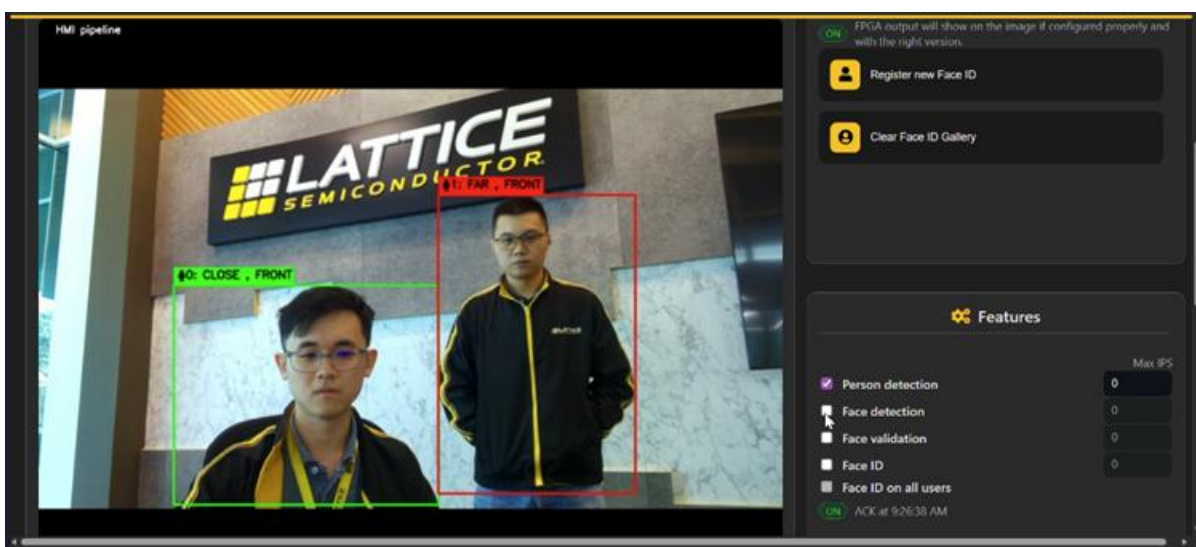


Figure 4.6. Pipeline Control with Only Person Detection

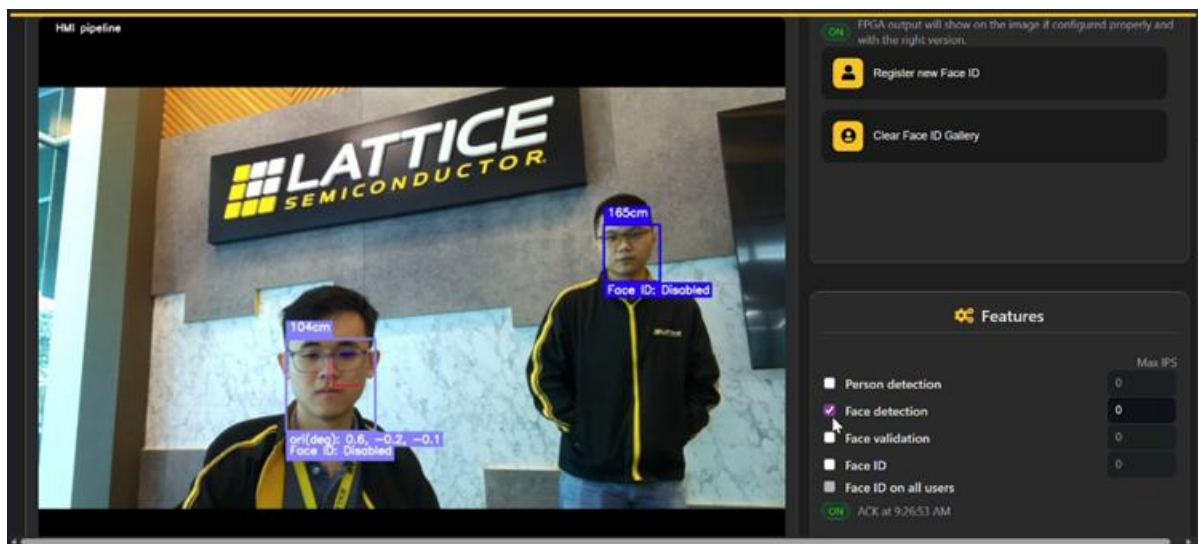


Figure 4.7. Pipeline Control with Only Face Detection

4.2.3. Maximum IPS

The *Max IPS* setting sets the maximum number of times the feature can run per second. The [AI Pipeline](#) section describes the pipeline operation.

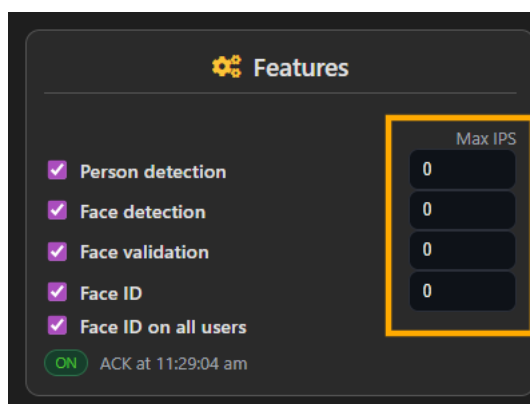


Figure 4.8. Maximum IPS

4.2.4. Face ID on All Users

The *Face ID on all users* setting, when enabled, runs Face ID checks on everyone in Face Detection range and displays the results under all face boxes as shown in [Figure 4.10](#). If it is unchecked, the Face ID only runs on the main user as shown in [Figure 4.11](#).

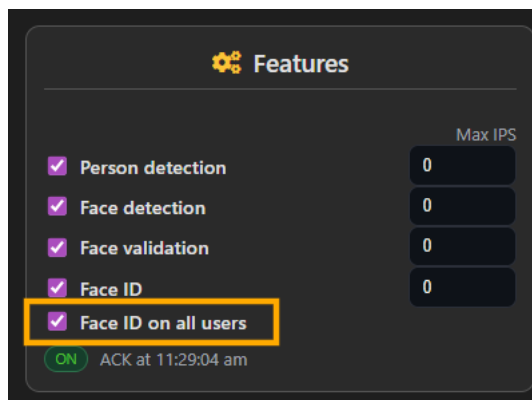


Figure 4.9. Face ID on All Users Setting in Pipeline Control

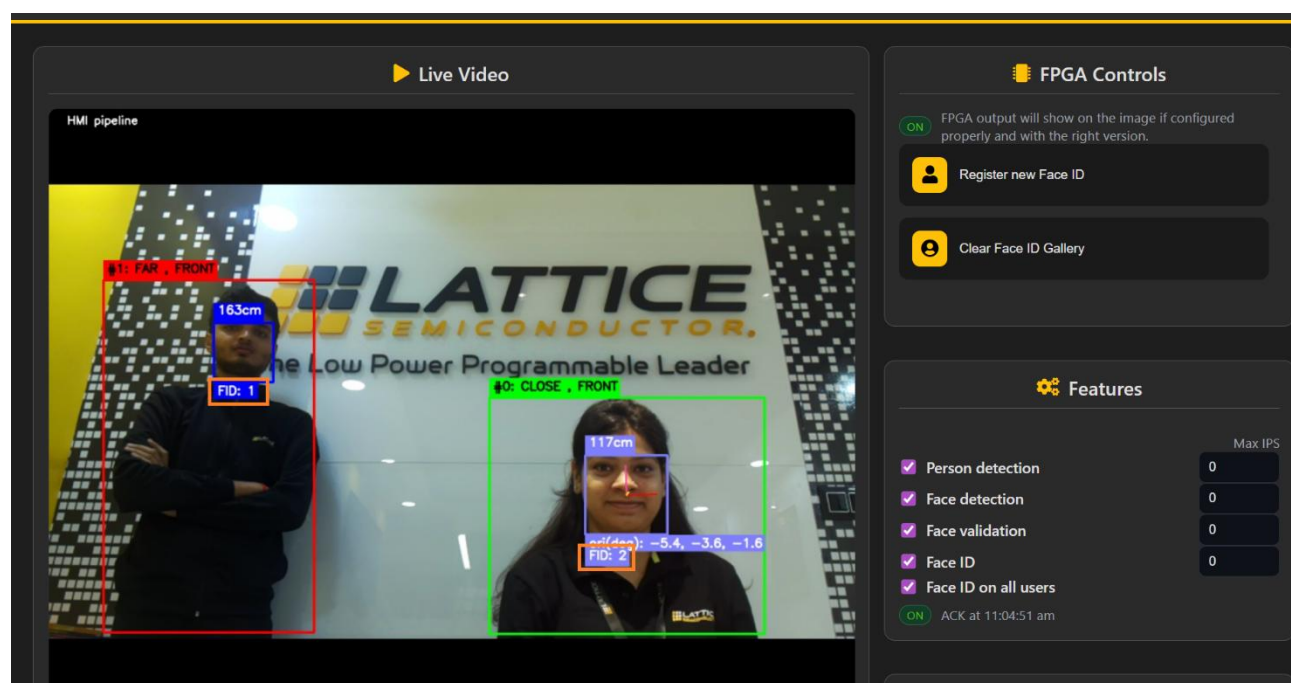


Figure 4.10. Face ID on All Users

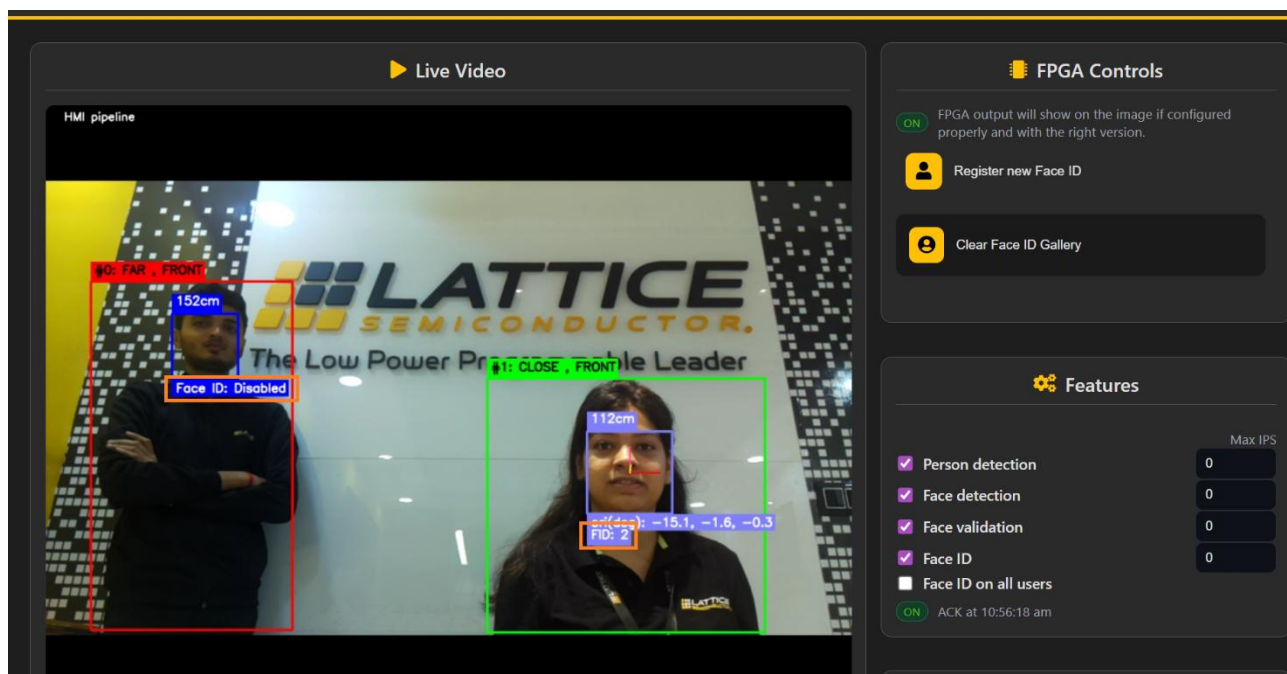


Figure 4.11. Face ID on Main User Only

4.2.5. Register New Face ID

The *Register new Face ID* button triggers a registration on the primary user denoted by the light blue face box. Subsequent Face ID checks return the user index in the gallery as shown in Figure 4.13.

Registration only works if the primary user is looking directly at the camera. If the requirement is not met, the registration is attempted until it meets the following conditions:

- Pitch > -20° and Pitch < 20°
- Yaw > -15° and Yaw < 15°
- Roll > -15° and Roll < 15°

In addition, the *Requirement Unmet* is labeled on the metadata screen shown in Figure 4.14.

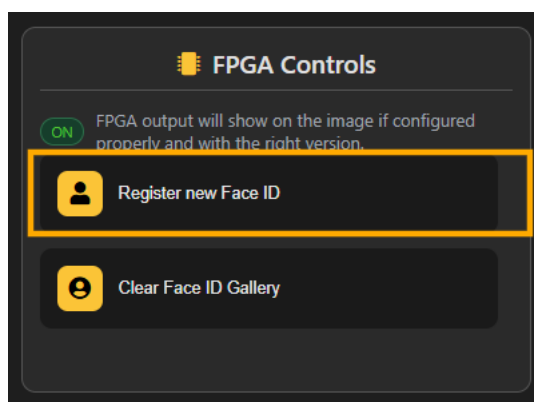


Figure 4.12. Register New Face ID



Figure 4.13. Registered Face ID



Figure 4.14. Face ID – Requirement Unmet

4.2.6. Clear Face ID Gallery

The *Clear Face ID Gallery* button removes everyone that registered from the gallery. After this, the result of Face ID must be *No gallery*.

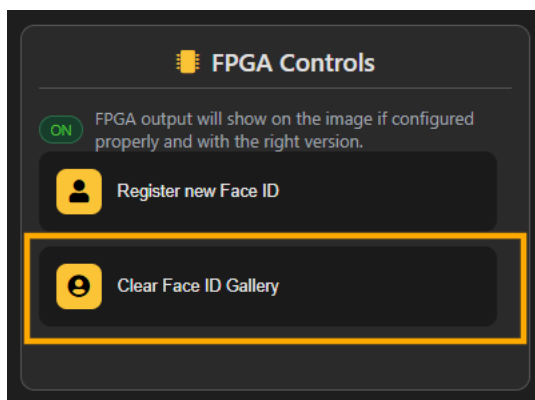


Figure 4.15. Clear Face ID Gallery

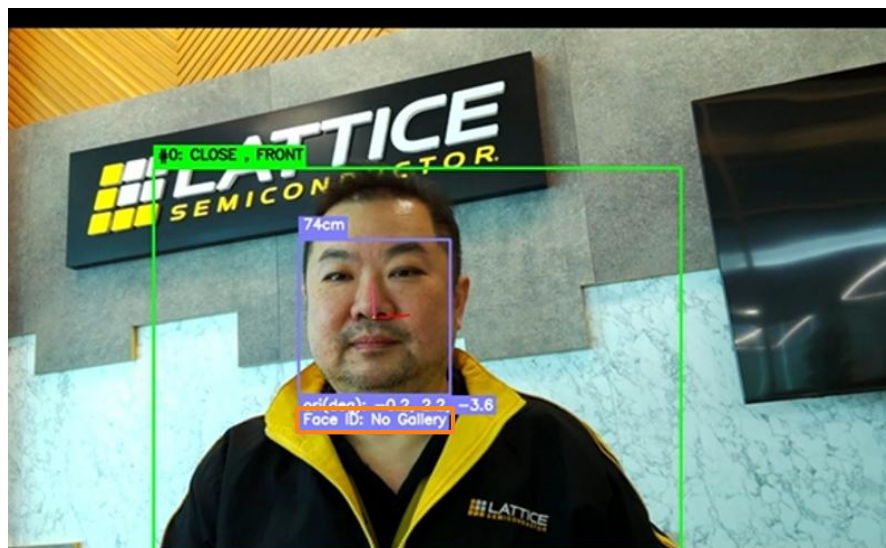


Figure 4.16. Face ID with No Gallery Shown After Clear

4.3. Metrics Screen

There are two sections – Software and Hardware Metrics.

- Hardware metrics – real time temperature and power sensor data
- Software metrics – real time throughput FPS

The right-aligned sidebar displays the system hardware metrics and can be opened using the menu icon in the top-right corner. On-chip sensors capture real-time telemetry and transmit this data to the upper software layers. The primary assembly consists of the CM5 and the SOM components; temperature telemetry is available specifically for the CM5, while power consumption readings are monitored for both the CM5 and the SOM (as shown in Figure 4.17).

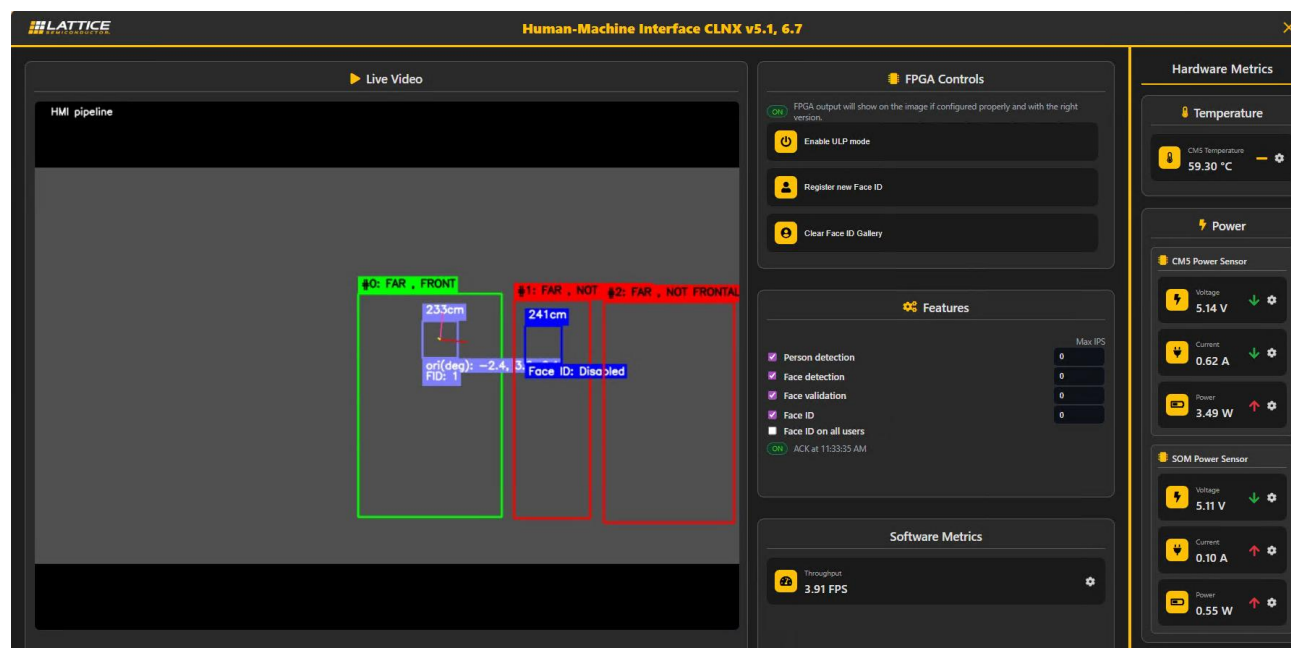


Figure 4.17. Hardware and Software Metrics

5. Running the HMI Demo Design on CrossLinkU-NX SoM HW Board

5.1. Requirements

The following lists the requirements for the design demonstration.

5.1.1. Hardware

Table 5.1. Available Hardware in the CrossLinkU-NX SoM Demo Kit (OPN: LIFCL-SOM-EVN)

Quantity	Description	Part Number/Manufacturer
1	CrossLinkU-NX SoM HW Board	PN: LIFCL-SOM-EVN Rev B, MFG: Lattice Semiconductor
1	Generic Carrier HW Board	PN: LF-GEN-CR-PCBA Rev A, MFG: Lattice Semiconductor
1	Raspberry Pi CM5	PN: SC1601, MFG: Raspberry Pi
1	Raspberry Pi CM5 Passive Cooler	PN: SC1752, MFG: Raspberry Pi
1	IMX219 Camera with regular (narrow) FOV	PN: (ArduCam B0191), MFG: Arducam
1	22-22 RPi 5 Camera Cable 200 mm	PN: B0BQ72Q33C, MFG: SHIVLAHERI
1	USB Cable Type A Male to Type C Male 3.00'	PN: U428-003-G2, MFG: Eaton Tripp Lite
1	IMX219 Camera holder	PN: R213343 (web store SKU), MFG: KKSBB
1	5 V 25 W AC/DC External Wall Mount (Class II) Adapter Multi-Blade (Sold Separately) Input	PN: WR9QA5000USBCNAN(R6B), MFG: GlobTek, Inc.
1	Input Plug for Wall Adapter	PN: Q-KIT®, MFG: GlobTek, Inc.
1	Modular Cable Plug to Plug 8p8c (RJ45, Ethernet) 3.28' (1.00 m) Shielded	PN: BC-5SG010M, MFG: Bel Inc.
1	USB 2.0 Cable Type A Male to Micro B Male 3.28' (1.00 m) Shielded	PN: CBL-UA-MUB-1 MFG: Harwin Inc.
3	Conn Shunt 1.27 mm black handle – Used to enable eMMC write mode on the CM5	PN: M50-2000005 MFG: Harwin Inc.

5.1.2. Software

The pre-installed CrossLinkU-NX SoM HMI demo package includes:

- Raspberry Pi OS – Bookworm
- Lattice EVE SDK
- Lattice Web Server

Below is the demo bring-up utility:

- [PuTTY](#)

The following are the system recovery tools:

- [Lattice Radiant Programmer](#) 2025.1 and above
- [Raspberry Pi Imager](#) v1.9.6 and above
- [Raspberry rpiboot](#)

5.2. Board Setup

The following sections show the available boards.

5.2.1. CrossLinkU-NX SoM HW Board

The CrossLinkU-NX SoM HW board (LIFCL-33U-SOM-EVN) is a compact SoM HW board that handles the MIPI CSI-2 camera input, AI inference pipeline, and low-power management. It includes connectors for Camera Sensor (*J6*), USB-C (*J3*), header pins for JTAG programming (*J7*-Reserved) and board to board connector to generic carrier HW board (*J1* and *J2*).

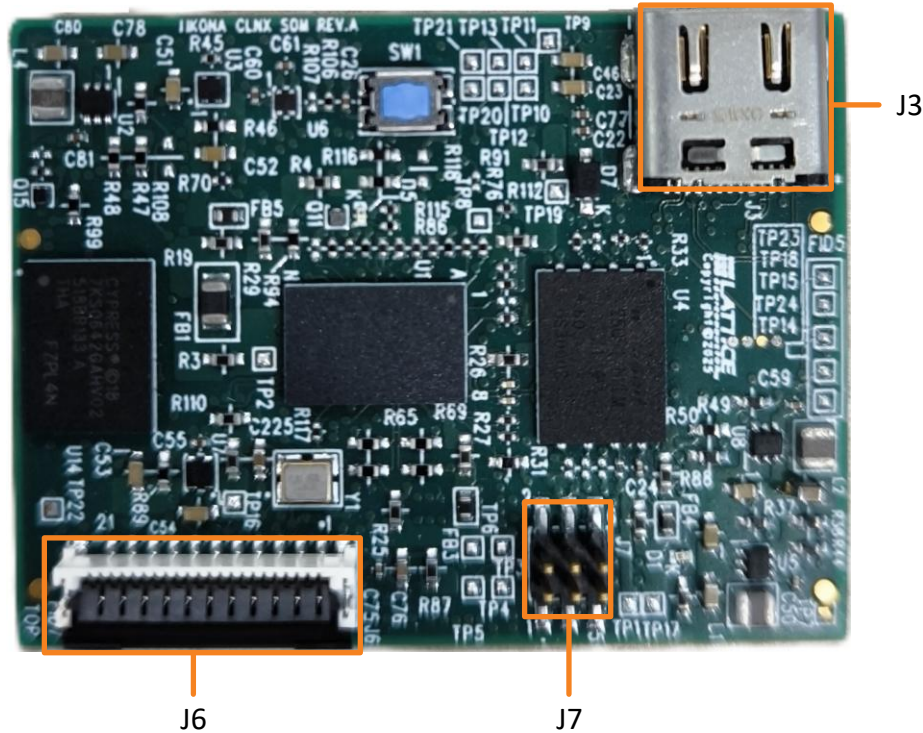


Figure 5.1. CrossLinkU-NX SoM HW Board (Top View)

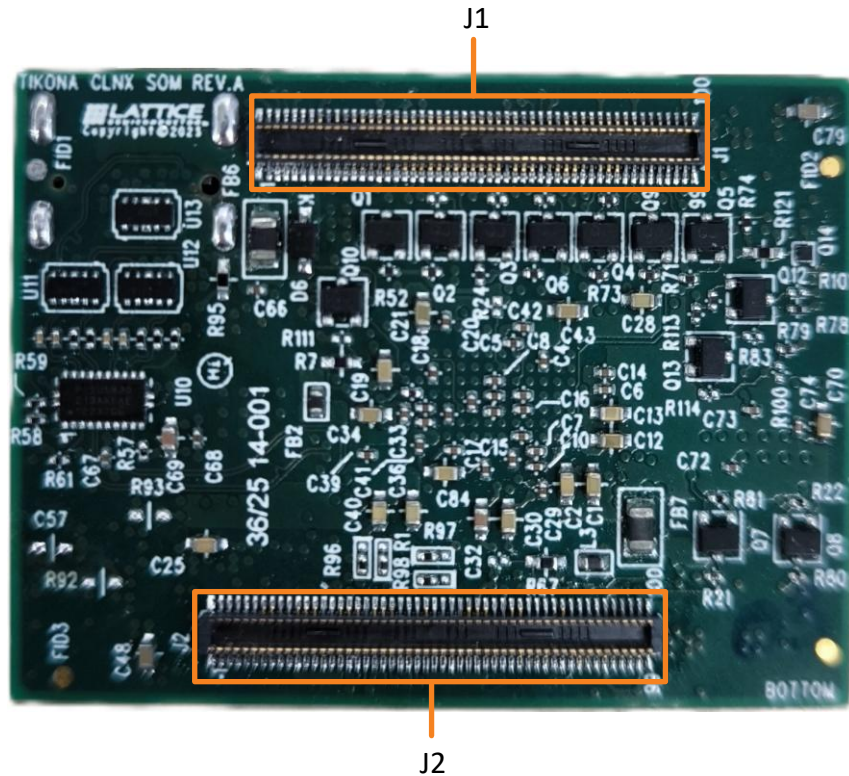


Figure 5.2. CrossLinkU-NX SoM HW Board (Bottom View)

5.2.2. Generic Carrier HW Board

The Generic Carrier HW board (LF-GEN-CR-PCBA) acts as a central interface, bridging the CrossLinkU-NX SoM and CM5. The key connections include:

- Power Connector, USB and Ethernet Ports: The carrier board routes the USB Type C (*J19*) for power and host communication, USB Type A (*J13*) for firmware update and RJ45 Ethernet connections (*J1*) for network access.
- UART and JTAG Interfaces: FTDI controller on the carrier board provide UART console access for CM5 and JTAG configurations for CrossLinkU-NX SoM. It is accessed through USB Micro Type B port (*J16*).
- Board to board connectors: One pair for interfacing with CrossLinkU-NX SoM board (*J14* and *J15*). The second pair for interfacing Raspberry Pi CM5 (*J6* and *J7*).

WiFi option is disabled by default.

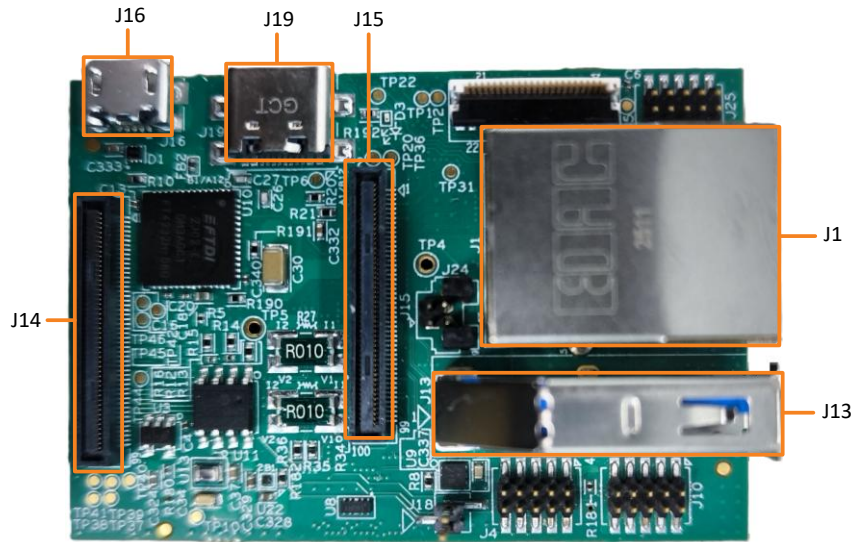


Figure 5.3. Generic Carrier HW Board (Top View)

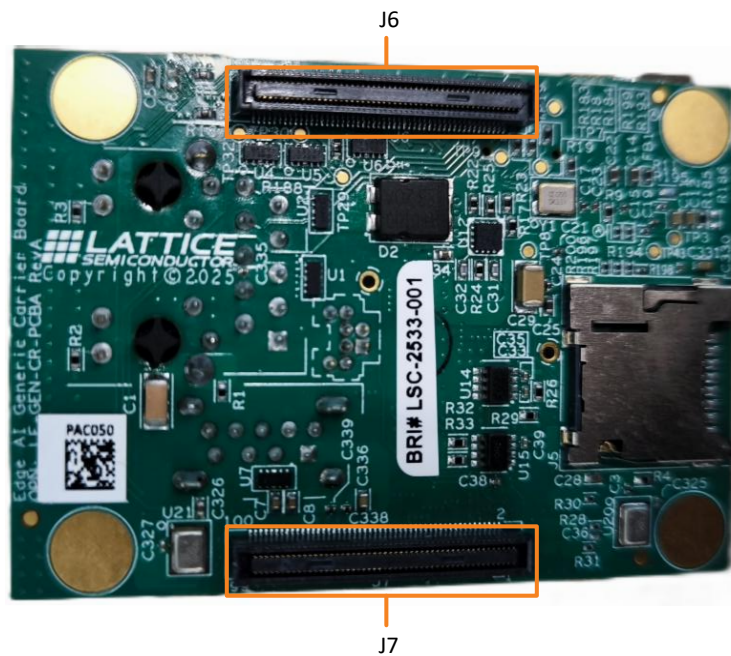


Figure 5.4. Generic Carrier HW Board (Bottom View)

5.2.3. Raspberry Pi CM5

The host compute module running the Raspberry Pi OS handles the AI metadata processing with EVE SDK and provides visualization dashboard through web server. The host module is equipped with USB, Ethernet, and GPIO interfaces.



Figure 5.5. Raspberry Pi CM5 (Top View)



Figure 5.6. Raspberry Pi CM5 (Bottom View)

5.2.4. IMX219 Camera Module (Narrow FOV – ArduCam B0191)

The IMX219 Camera Module is a MIPI CSI-2 camera sensor that supports RAW10 format. This is used for video streaming and AI inferencing.



Figure 5.7. IMX219 Camera Module (Top View)



Figure 5.8. IMX219 Camera Module (Bottom View)

5.3. Board-to-Board Connections

This section describes the physical interconnections between the CrossLinkU-NX SoM HW board, the generic carrier HW board, Raspberry Pi CM5, and the camera module. Proper board-to-board connectivity is essential to ensure reliable data transfer for MIPI streaming and AI inferencing.

Figure 5.9 shows the physical stacking and orientation of the three main hardware boards used in the CrossLinkU-NX SoM HMI demonstration design.

To connect the board, perform the following:

1. From the right side of Figure 5.9, mount the **CrossLinkU-NX SoM HW board** onto the **Generic Carrier HW board** using the designated board-to-board connectors.
2. Mount the **Generic Carrier HW board** (with the **CrossLinkU-NX SoM HW board** attached) onto the **RPi CM5**.

All boards must be stacked vertically without changing orientation as depicted in Figure 5.9, ensuring that all board-to-board connectors are fully engaged to provide secure electrical and mechanical integration.

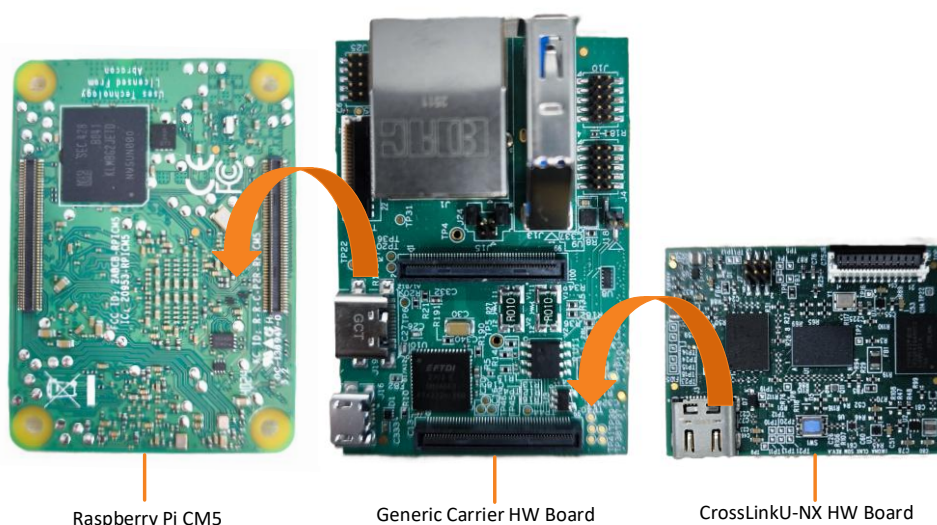


Figure 5.9. Stacking Orientation Arrangement

Figure 5.10 shows the fully assembled stack of hardware boards. Before proceeding, verify that all boards are correctly aligned and firmly seated in their respective connectors to ensure secure electrical and mechanical integration.

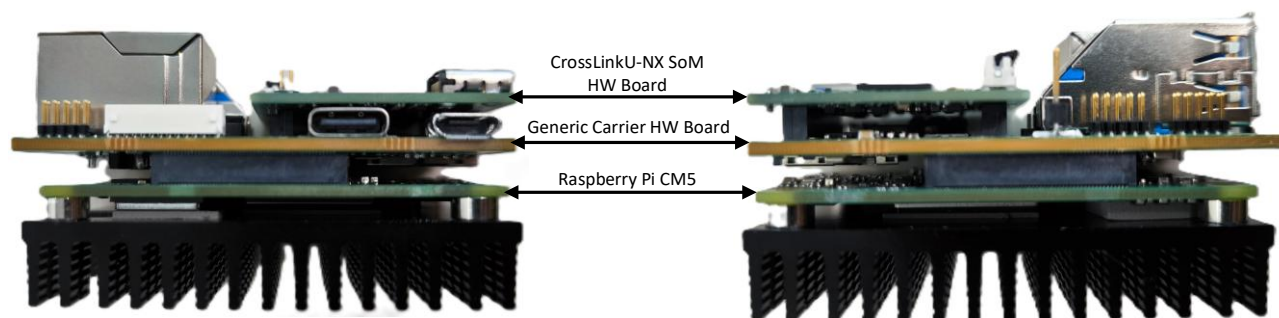


Figure 5.10. Board-to-Board Connections

The camera module connects to the CrossLinkU-NX SoM HW board through an FFC cable. Ensure the cable's contact is correctly aligned with the MIPI CSI-2 connector. The blue stiffener side must face the connector latch, as shown in [Figure 5.11](#). To secure the cable, lift the connector latch, insert the cable fully and press the latch down to lock it in place. Avoid sharp bends in the FFC cable as it is fragile and may be damaged by excessive stress. Before powering the system, verify all connections to prevent damage or communication errors.

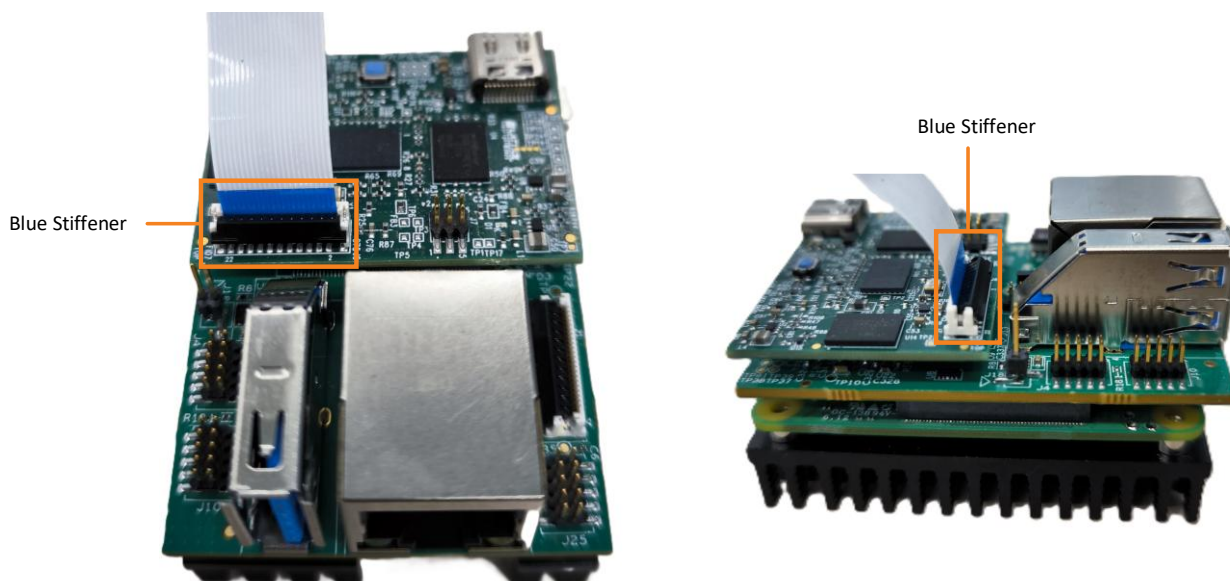


Figure 5.11. Board Connections with Camera FFC Cable

5.4. Configuring the Hardware Jumper

To configure the hardware jumper, connect the pin header 3 and 7 and 6 and 8 on the J24 jumper as [Figure 5.12](#). This allows the RPi to receive UART from the CrossLinkU-NX SoM.

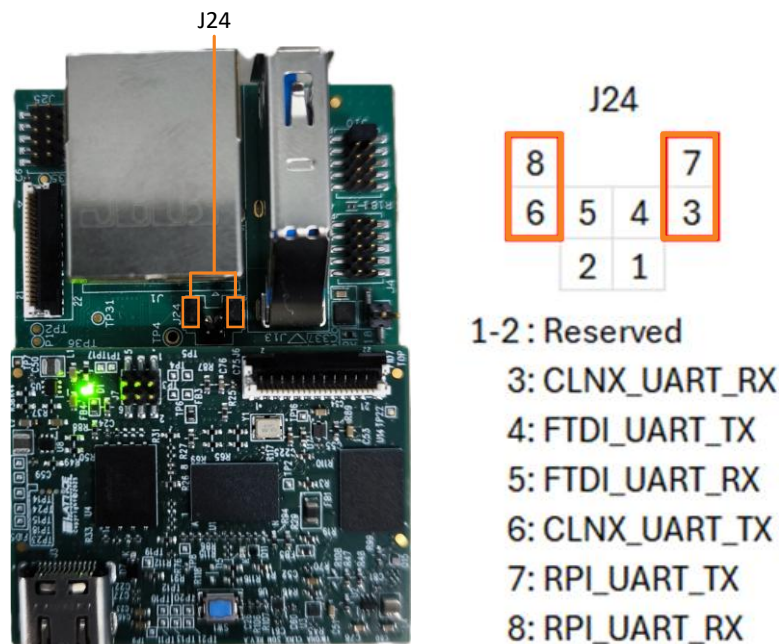


Figure 5.12. Hardware Connections for UART

5.5. RPi CM5 Serial Console Access

This section explains how to access Raspberry Pi CM5's Linux terminal through its built-in serial console. This allows for network-free headless access. The minimal connection required for serial console access is shown in Figure 5.13. A USB Type C power brick provides power to the module while a USB Type A to Micro Type B cable provides connectivity from the Host PC to Raspberry Pi CM5.

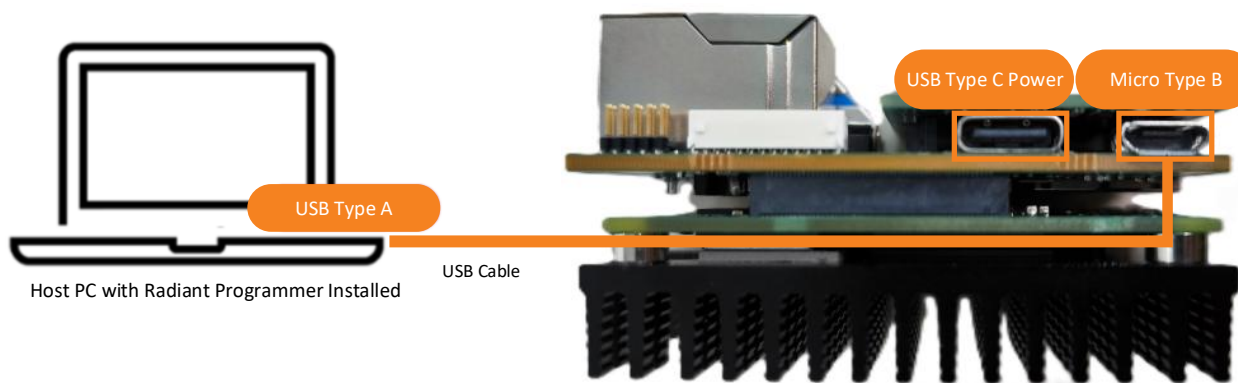


Figure 5.13. Hardware Setup for Serial Console Access

To access the Raspberry Pi console terminal, perform the following:

1. Connect the board to the PC using a USB Type A to Micro Type B cable, as shown in Figure 5.13. Connect the power adapter to the Raspberry Pi CM5 board and power on the adapter.
2. After powering on the module, the required COM port for serial console access is shown in the host PC's device manager. There are four new COM ports available to select. Open each of the *Properties* window to get the location name of **on USB Serial Converter D**.

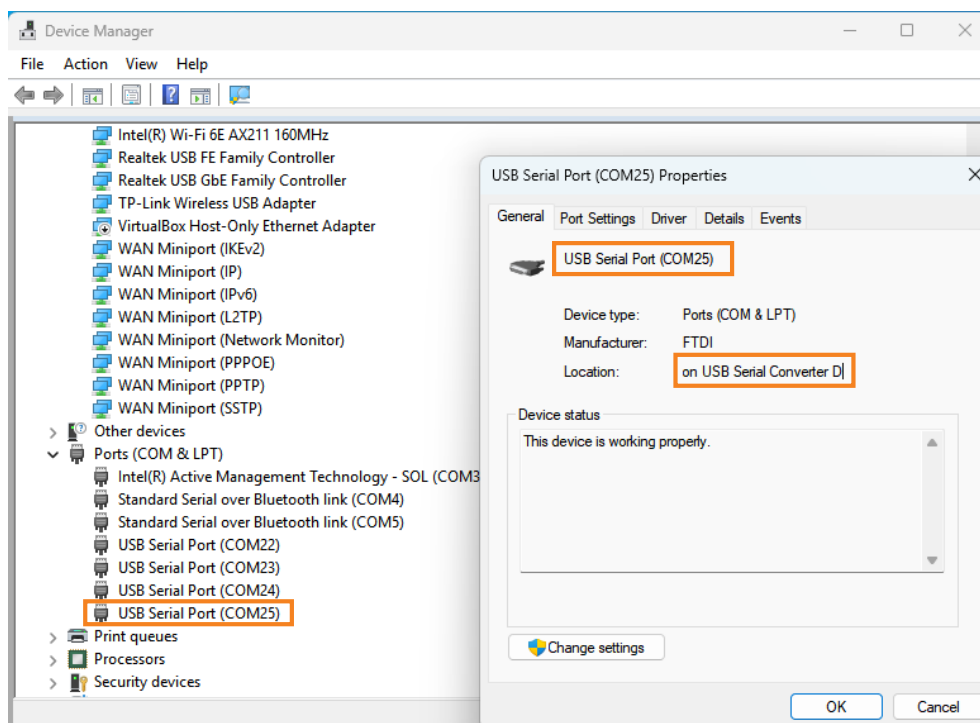


Figure 5.14. Available COM Ports in Device Manager

- Follow the steps as shown in Figure 5.15 to open the selected COM port with PuTTY. COM settings for PuTTY are stated below.

Baud Rate: 115200
Data Bits: 8
Parity: None
Stop Bits: 1
Flow Control: None

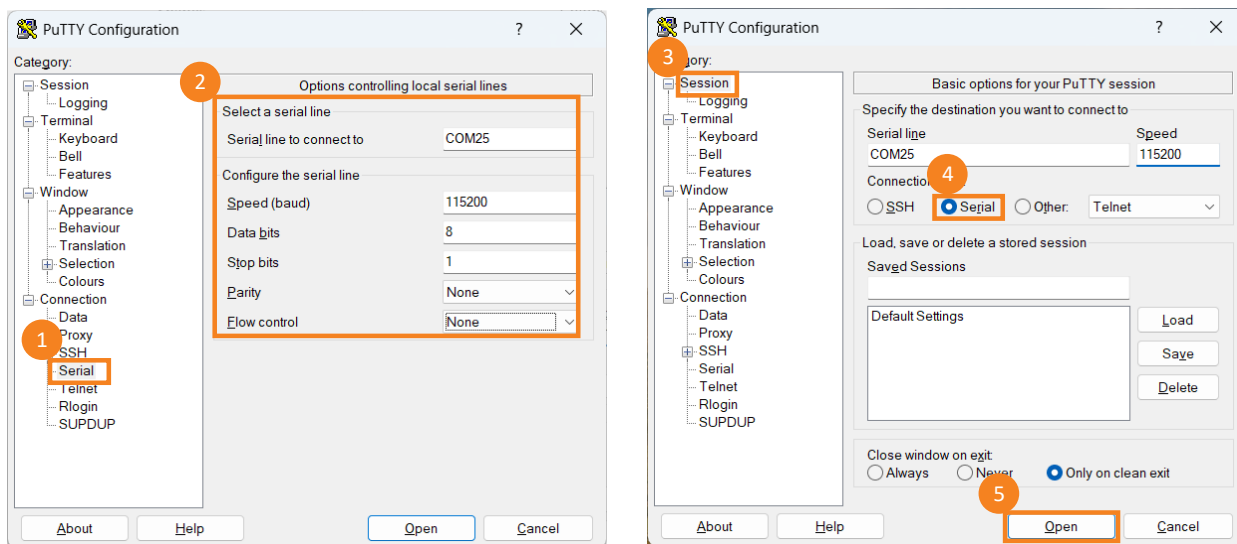


Figure 5.15. PuTTY Configuration to Open COM Port

- After opening serial port connection, there must be a blank screen only. Press **Enter** to send a character and the login prompt is displayed as shown in [Figure 5.16](#).

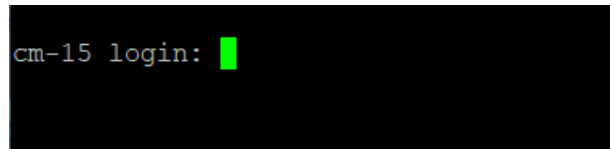


Figure 5.16. RPi CM5 Login Screen

- Log in with the default credentials. A terminal prompt is displayed as shown in [Figure 5.17](#).

Username: lattice
Password: lattice

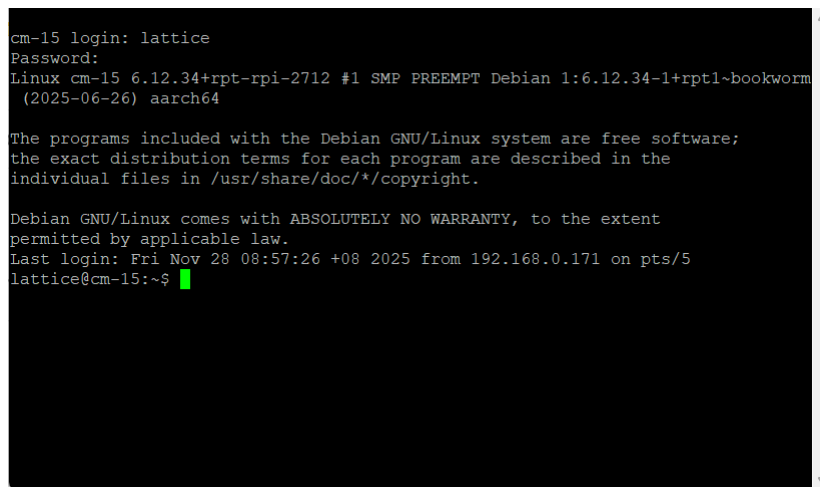


Figure 5.17. Serial Terminal Prompt

5.6. Connecting the Raspberry Pi CM5 Network

The HUB Raspberry Pi OS image comes pre-installed with an ISC DHCP Server for serving dynamic IP addresses to devices connected to the Ethernet port (eth0). The DHCP server starts automatically on boot with a static IP address.

To set up and use the Ethernet connection for the SOM Raspberry Pi CM5 after flashing:

- Your host machine should have an Ethernet port with a dynamic IP address configured.
- Connect an Ethernet LAN cable from your machine to the CrossLinkU-NX SoM kit's LAN connection (as shown in [Figure 5.18](#)).
- Connect the power supply to the SOM.
- Optionally, use PuTTY to access the serial debug UART and confirm the IP address.

Note: For terminal access, refer to the [RPi CM5 Serial Console Access](#) section.

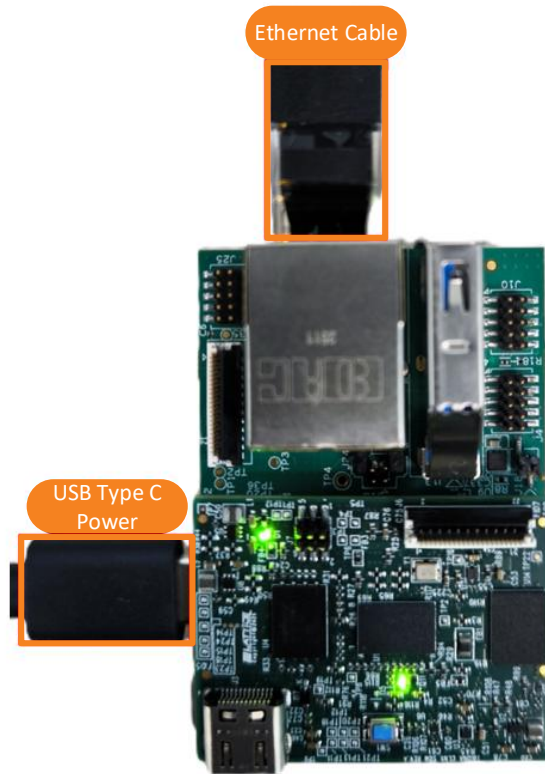


Figure 5.18. CrossLinkU-NX SoM with Ethernet Connection

A static IP address `10.10.1.1/24` is automatically assigned to the SOM Raspberry Pi CM5. Refer to [Figure 5.19](#) to verify that the assigned IP address is accessible through SSH.

If this IP address is pre-allotted, a subsequent IP in the range `10.10.1.100` to `10.10.1.200` is assigned. Connect to the CM5 terminal through a serial connection and run the command below to obtain the IP address of the RPi CM5:

```
hostname -I
```

```
lattice@lattice-hub:~ $ hostname -I
10.10.1.1
lattice@lattice-hub:~ $
```

Figure 5.19. Check Host IP Command

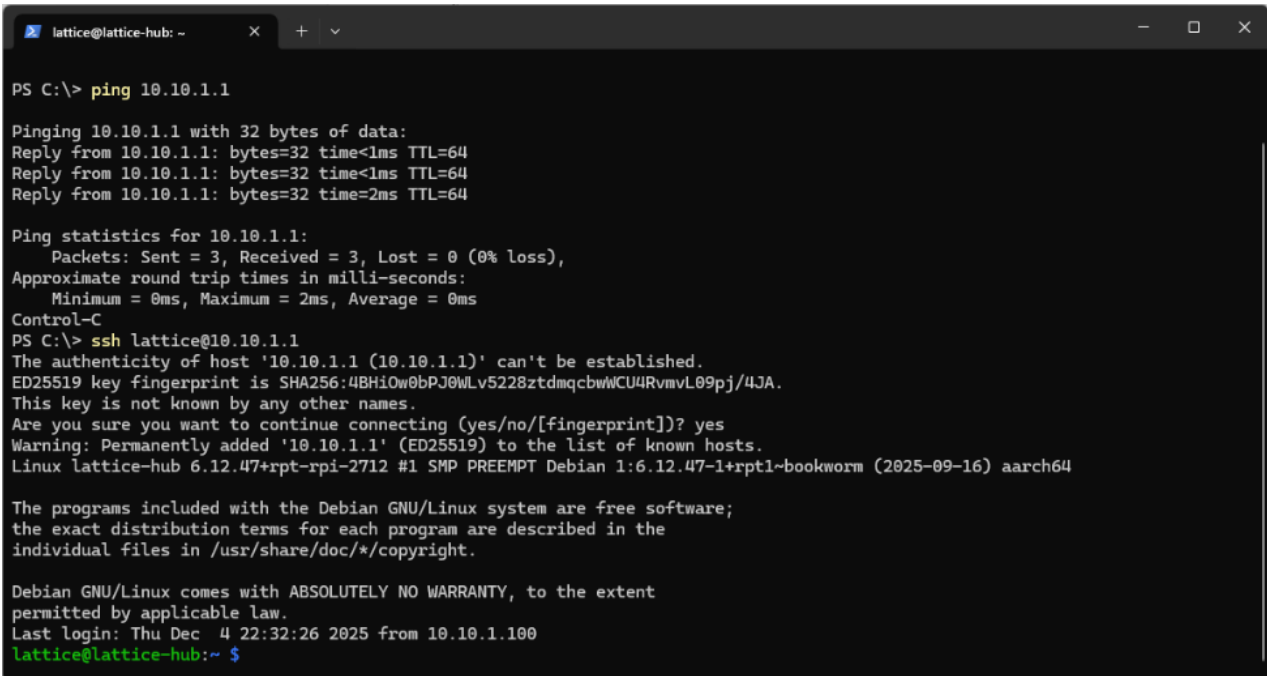
5.6.1. RPi CM5 Terminal Access through SSH

The IP address is now accessible within the network. In case of failure, power cycle the SOM for a seamless reboot and IP allocation. Run the following command to access the CM5 terminal over the network. Once connected, the serial cable is no longer required.

```
ssh lattice@10.10.1.1
```

The following are the default credentials to access the SOM assembly over SSH:

- Username: lattice
- Password: lattice



```
lattice@lattice-hub: ~
PS C:\> ping 10.10.1.1

Pinging 10.10.1.1 with 32 bytes of data:
Reply from 10.10.1.1: bytes=32 time<1ms TTL=64
Reply from 10.10.1.1: bytes=32 time<1ms TTL=64
Reply from 10.10.1.1: bytes=32 time=2ms TTL=64

Ping statistics for 10.10.1.1:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
Control-C
PS C:\> ssh lattice@10.10.1.1
The authenticity of host '10.10.1.1 (10.10.1.1)' can't be established.
ED25519 key fingerprint is SHA256:4BHi0w0bPJ0wLv5228ztdmqcbwWCU4RvmvL09pj/4JA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.1.1' (ED25519) to the list of known hosts.
Linux lattice-hub 6.12.47+rpt-rpi-2712 #1 SMP PREEMPT Debian 1:6.12.47-1+rpt1~bookworm (2025-09-16) aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Dec  4 22:32:26 2025 from 10.10.1.100
lattice@lattice-hub:~ $
```

Figure 5.20. Windows Terminal Accessing SOM Through SSH Over Ethernet

5.6.2. Enabling Static IPv4 Address (Optional)

This configuration enables the SOM to be accessed over the LAN network with a fixed address. To set a static IP for the Raspberry Pi CM5 on the SOM assembly, run the following command. This needs to be configured only once and it automatically reconnects if available:

```
sudo nmcli connection modify <connection-name> ifname eth0 type ethernet ipv4.addresses <ip-addr>/24 ipv4.gateway <gateway> ipv4.method manual autoconnect yes
```

Where:

- <connection-name> – Wired connection 1 (already set up on Raspberry Pi CM5)
- <ip-addr> – IPv4 address
- <gateway> – IPv4 gateway

Note: Wi-Fi is disabled by default on the generic carrier board. Root privileges are required to modify network connections. If you encounter a permission denied error, ensure you run the command using sudo or with a user account that has elevated privileges.

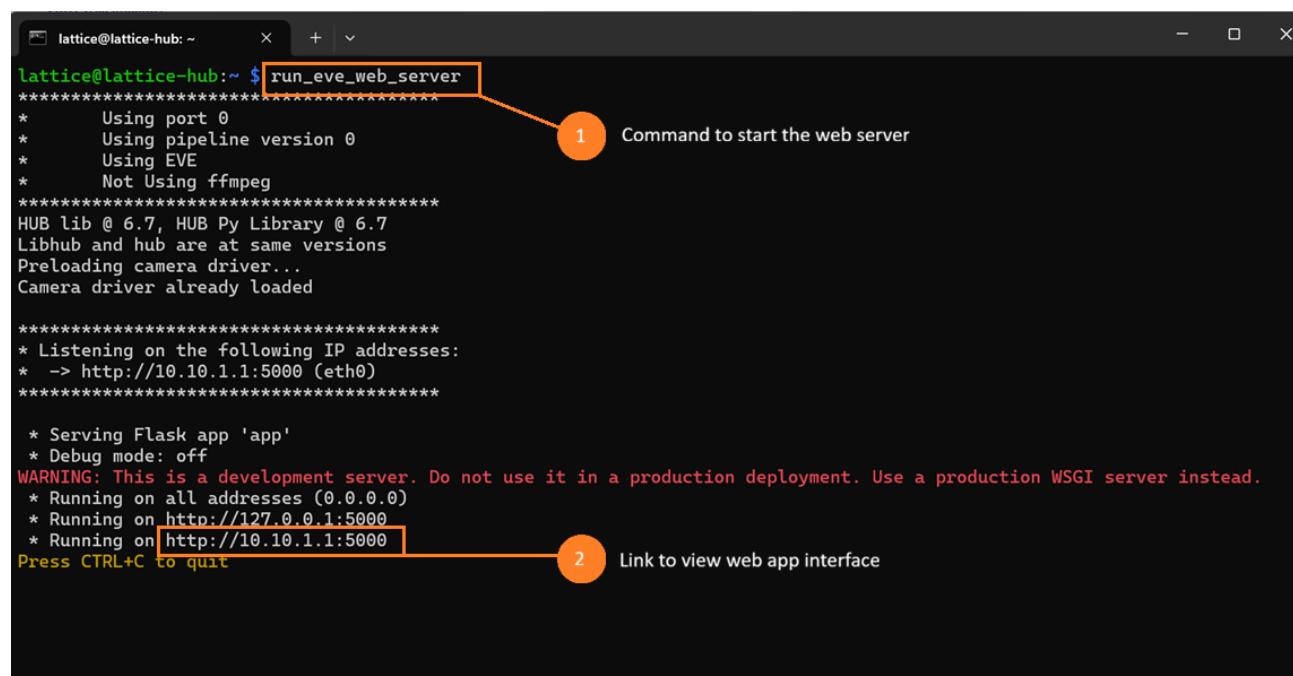
5.7. Running EVE and Web Server

To run the EVE and web server, perform the following:

1. For terminal access, refer to the [RPI CM5 Terminal Access through SSH](#) section.
2. Start the web server.

```
run_eve_web_server
```

Note the **web app interface URL** in the logs as shown in [Figure 5.21](#).



```
lattice@lattice-hub: ~  
lattice@lattice-hub:~$ run_eve_web_server  
*****  
* Using port 0  
* Using pipeline version 0  
* Using EVE  
* Not Using ffmpeg  
*****  
HUB lib @ 6.7, HUB Py Library @ 6.7  
Libhub and hub are at same versions  
Preloading camera driver...  
Camera driver already loaded  
  
*****  
* Listening on the following IP addresses:  
* -> http://10.10.1.1:5000 (eth0)  
*****  
  
* Serving Flask app 'app'  
* Debug mode: off  
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.  
* Running on all addresses (0.0.0.0)  
* Running on http://127.0.0.1:5000  
* Running on http://10.10.1.1:5000  
Press CTRL+C to quit
```

1 Command to start the web server

2 Link to view web app interface

Figure 5.21. IP Address of Web Server

3. In your host computer, open a web browser and enter the address obtained in Step 2. For details on navigating the Web App interface, refer to the [Web Server](#) section.

Notes:

1. Terminate the Web Server: After completing the operation, terminate the web server by pressing CTRL+C in the terminal where the application was launched.
2. Switching Modes: To switch between streaming and sensing modes, first terminate the web server. Then, relaunch the web server to return to the landing page.

6. Running the Firmware Management Application

Before executing the application, make sure the web server is not running.

6.1. Updating the Firmware

To update the firmware on the FPGA, perform the following steps:

1. Connect the USB Type C to Type A cable from USB Type C connector (J3) on the CrossLinkU-NX SoM HW board to USB Type A connector (J13) on the carrier board. This connects RPi CM5's USB to CrossLinkU-NX SoM as shown in [Figure 6.1](#).

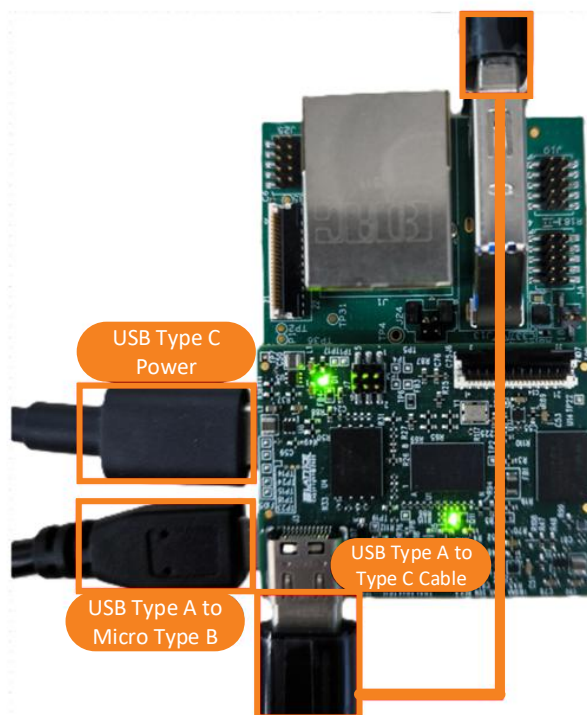


Figure 6.1. Hardware Connections for Firmware Update

2. For terminal access, refer to the [RPi CM5 Serial Console Access](#) section.
3. Navigate to the **FW_UPDATE_TOOL** folder (Path: `/opt/clnx_som/FW_Update`) to access the Lattice Firmware Management Application.
4. Run the executable.

```
sudo ./clnxsomdemo_fwupd_<firmware_version>
```

Example: `sudo ./clnxsomdemo_fwupd_01.12.00.00`
5. Wait for the update to be completed. The executables display the progress in the terminal. Do not disconnect the device or power off during this step.
6. Once the update is completed, it displays the success message shown in [Figure 6.2](#). Power cycle the board after programming.

```
2025-12-04 14:55:33,524 [INFO] [FwUpd 01.02.00.00] Starting SoM Firmware Update to 01.12.00.00
2025-12-04 14:55:34,106 [INFO] Updating Current NX [5403] to Target [5405]
2025-12-04 14:55:52,251 [INFO] Skipping update for FM (Version: 5303)
2025-12-04 14:55:52,251 [INFO] Updating Current RISCv [5506] to Target [550F]
2025-12-04 14:55:57,421 [INFO] Skipping update for FD (Version: 0115)
2025-12-04 14:55:57,422 [INFO] Skipping update for FID (Version: 0511)
2025-12-04 14:55:57,423 [INFO] Skipping update for LV (Version: 0412)
2025-12-04 14:55:57,423 [INFO] Skipping update for PARSEr (Version: 0903)
2025-12-04 14:55:57,424 [INFO] Updating Current PD [0612] to Target [0614]
2025-12-04 14:56:13,010 [INFO] Update completed successfully.
```

Figure 6.2. Firmware Update Success Message

6.2. Firmware Version

The following steps queries the bitstream version currently on the hardware device.

1. Follow steps 1 to 3 in the [Updating the Firmware](#) section.
2. Run the executable.

```
sudo ./clnxsomdemo_fwupd_<firmware_version> -v
Example: sudo ./clnxsomdemo_fwupd_01.12.00.00 -v
```

3. The executables display each of the bitstream versions in the terminal. Wait 30 seconds before running the web server.

```
2025-12-04 14:56:22,310 [INFO] =====
2025-12-04 14:56:22,310 [INFO] CLNX SoM Firmware Update Tool
2025-12-04 14:56:22,310 [INFO] Version : 01.02.00.00
2025-12-04 14:56:22,310 [INFO] =====
2025-12-04 14:56:22,310 [INFO] Global Version [01120000]
2025-12-04 14:56:22,310 [INFO] NX [5405]
2025-12-04 14:56:22,310 [INFO] FM [5303]
2025-12-04 14:56:22,310 [INFO] RISCv [550F]
2025-12-04 14:56:22,311 [INFO] FD [0115]
2025-12-04 14:56:22,311 [INFO] FID [0511]
2025-12-04 14:56:22,311 [INFO] LV [0412]
2025-12-04 14:56:22,311 [INFO] PARSEr [0903]
2025-12-04 14:56:22,311 [INFO] PD [0614]
```

Figure 6.3. Firmware Versions

7. Debug Methodology

This section outlines the recommended steps for debugging the design. By following this section, system functionality can be verified and potential issues during integration and testing can be identified.

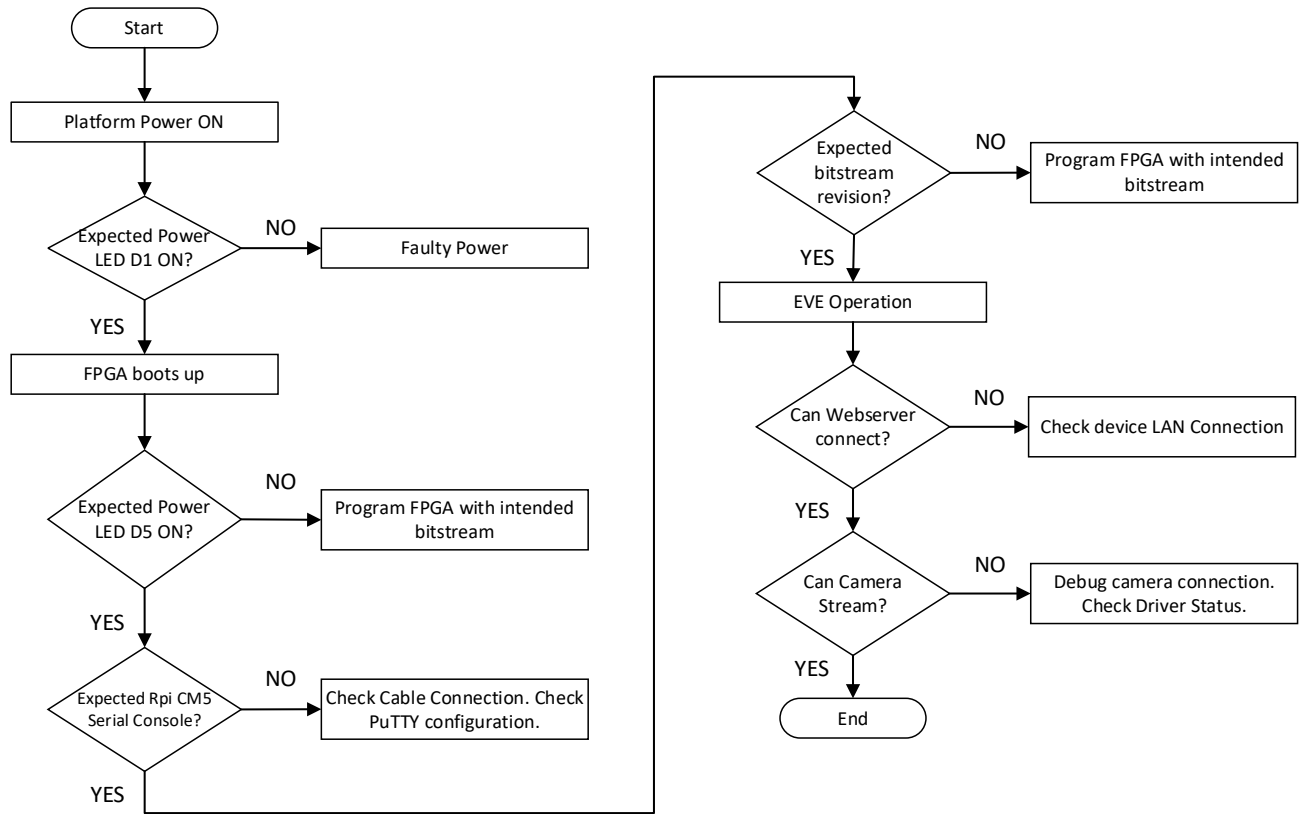


Figure 7.1. Debug Methodology Flow Chart

7.1. Powering Up the CrossLinkU-NX SoM

To power up the CrossLinkU-NX SoM, perform the following steps:

1. Power cycle the board.
2. This power cycle ensures the FPGA correctly loads and configures the new firmware.
3. Verify LEDs status.
4. Upon powering up, check the status of LEDs on the CrossLinkU-NX SoM HW board. Refer to [Figure 7.2](#) for LED locations. These LEDs indicate the board's operation status. [Table 7.1](#) describes each LED behavior upon boot up.

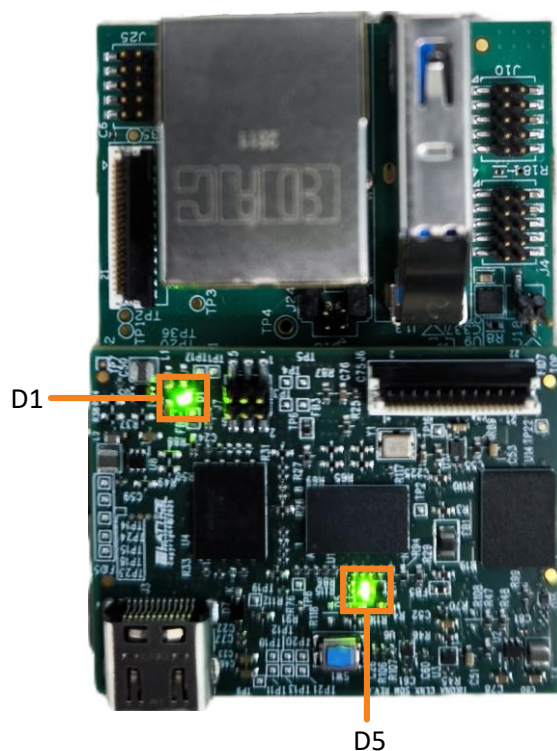


Figure 7.2. LED Position on CrossLinkU-NX SoM

Table 7.1. LED Status Upon Board Power Up

LED	Connected Signal	Board Power UP
D1	BOARD VCC Supply	ON
D5	CLNX_DONE	ON

7.1.1. Troubleshooting Common Issues

- Issue A: No LEDs are on.
 - Action 1: Check that the power supply for USB Type C cable is connected to the carrier board as shown in [Figure 5.13](#) and turn on.
- Issue B: LED D5 is not on.
 - Action 1: Reprogram the FPGA per the [Programming the Bit File](#) section to confirm the bitstream is updated.
 - Action 2: If the issue persists, submit a technical support case through www.latticesemi.com/techsupport.

7.1.2. Programming the Bit File

This section explains how to program a bitstream file to the CrossLinkU-NX System-on-Module (SoM). In most cases, this step is not required, as the CrossLinkU-NX SoM is pre-programmed with the HMI demonstration bitstream from the factory. You only need to perform these steps if you need to reprogram the module due to updates or recovery.

To program the bit file, perform the following:

1. Connect the board to the PC using a USB Type A to Micro Type B cable, as shown in [Figure 7.3](#). Connect the power adapter to the Raspberry Pi CM5 board and power on the adapter.

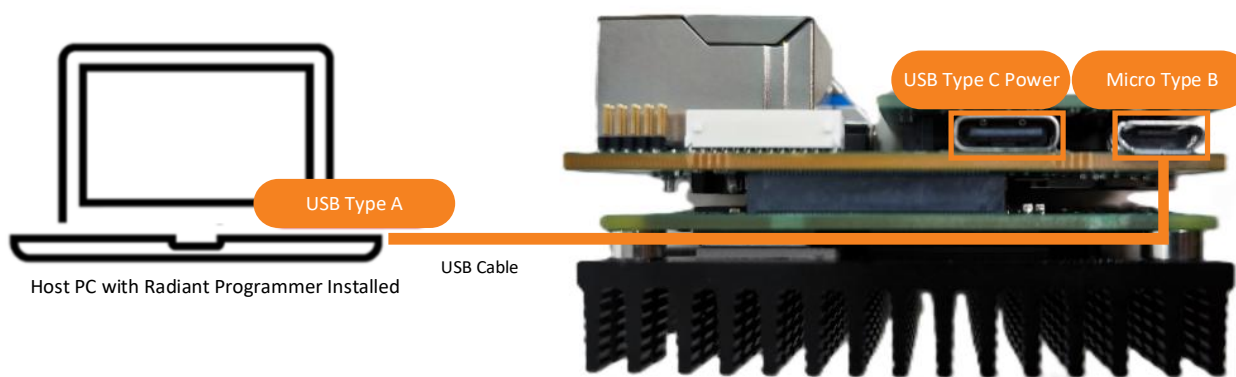


Figure 7.3. Hardware Setup for Bit File Programming

2. For terminal access, refer to the [RPI CM5 Serial Console Access](#) section.
3. Run the pre-programmer operation script.
`pre_pgmr_op`
4. Launch standalone Radiant Programmer on the Host PC. Enter the **Project Location** and **Project Name**; then click **OK**.

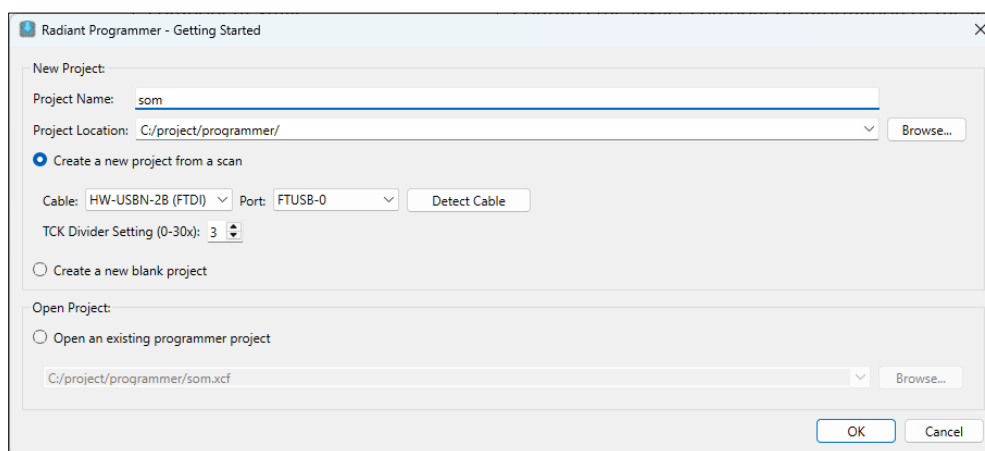


Figure 7.4. Radiant Programmer – Getting Started Window

5. Click **Detect Cable** in the cable setup section, select **FTUSB-0**, and then click **OK** to proceed.

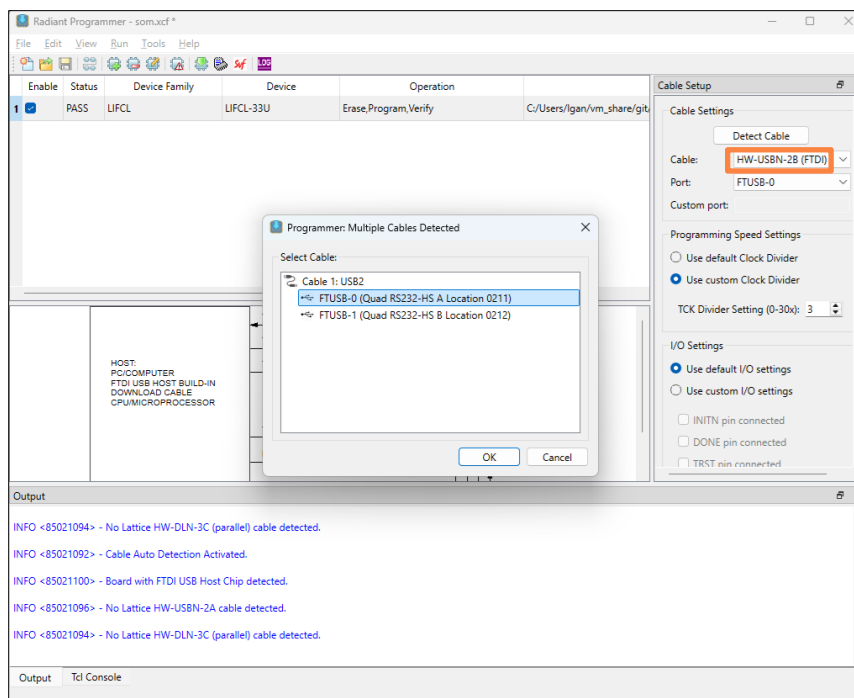


Figure 7.5. Radiant Programmer Detect Cable Settings

6. Double-click on **Erase, Program, Verify** in the **Operation** column to select the SPI Flash configuration and the bitstream file. The *Device Properties* window may take up to a minute to load.
7. Select the appropriate SPI flash configuration to flash directly to external SPI flash. Refer to Figure 7.6 for the expected settings. Click the ... button beside the file name bar and select the bitstream file from the *CLNX_SPI_IMAGE* folder listed in Table 2.1.

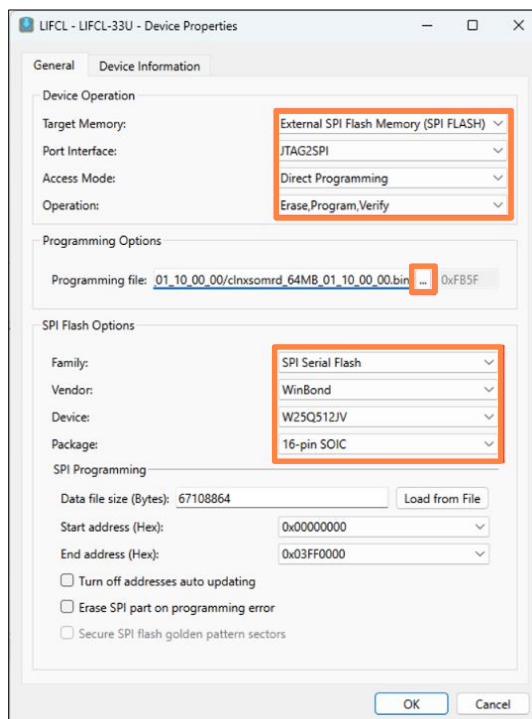


Figure 7.6 Selecting Bitstream File to Program

- Click the **Program Device** toolbar icon to program the bitstream into the board, as shown in [Figure 7.7](#). This process may take up to **20 minutes** to complete.



Figure 7.7 Program Device Toolbar Icon

- Verify that the programming status displays *Operation successful* in the output window, as shown in [Figure 7.8](#). To activate the flashed bitstream, power cycle the device.

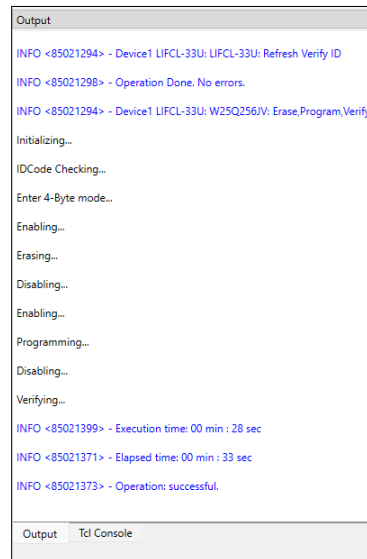


Figure 7.8. Message on Successful Programming

- For terminal access, refer to the [RPi CM5 Serial Console Access](#) section.
- Run the post-programmer operation script.

```
post_pgmr_op
```

7.2. Verifying Ethernet Connection

- For terminal access, refer to [RPi CM5 Serial Console Access](#).
 - On failure, check cable connection and PuTTY configuration.
 - If issue persists, refer to the [Flashing the RPi CM5](#) section to recover the RPi CM5's OS.
- To obtain the module's IP address, refer to the [Connecting the Raspberry Pi CM5 Network](#) section for the steps.
- Run command below on the host PC to check ethernet connectivity as shown in [Figure 7.9](#).

```
ping <IP_Address>
```

```
ping 192.168.0.236

Pinging 192.168.0.236 with 32 bytes of data:
Reply from 192.168.0.236: bytes=32 time=2ms TTL=64
Reply from 192.168.0.236: bytes=32 time=1ms TTL=64
Reply from 192.168.0.236: bytes=32 time=1ms TTL=64
Reply from 192.168.0.236: bytes=32 time=2ms TTL=64

Ping statistics for 192.168.0.236:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

Figure 7.9. Verify Raspberry Pi CM5 Ethernet Connectivity

7.3. Flashing the RPi CM5

This section explains the steps required to flash the operating system onto the Raspberry Pi Compute Module 5 (CM5). In most cases, this step is not required, as the CM5 typically comes with a factory-installed OS image. You can skip this section unless you need to re-flash the module due to corruption or update requirements.

To flash the RPi CM5, perform the following:

1. Connect pin header **1 and 2** on **Jumper J10** as shown in Figure 7.10. This shorts the nRPIBOOT and enables flashing of the RPi CM5.

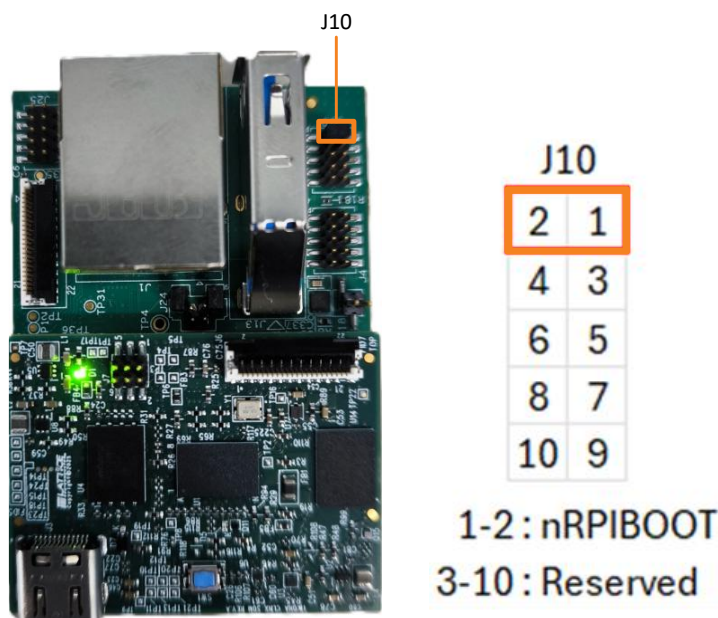


Figure 7.10. Hardware Connection for nRPIBOOT

2. Connect the board to the PC using a USB Type A to Type C cable as shown in Figure 7.11.

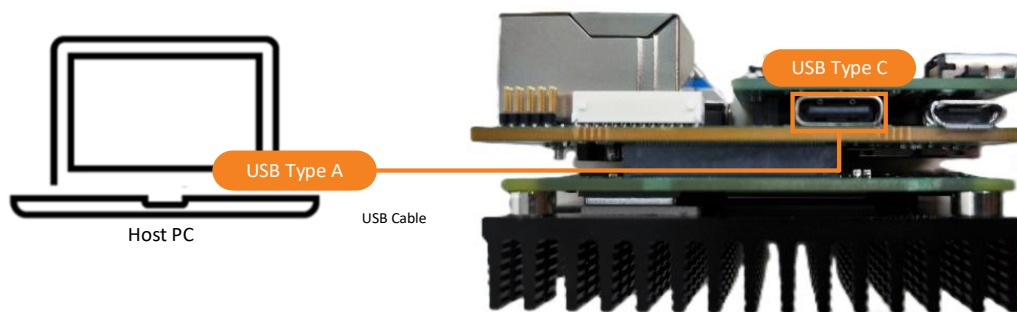


Figure 7.11. Hardware Setup for Flashing RPi CM5 OS

3. Run **rpiboot.exe** from the path `C:\Program Files (x86)\Raspberry Pi\rpiboot.exe`. This launches a Windows console, and if RPi CM5 is properly connected, the script continues to run and exits the terminal.

```

C:\Program Files (x86)\Raspbri  X  +  -  X
RPiBOOT: build-date 2025/05/19 pkg-version local 402baf02

Please fit the EMMC_DISABLE / nRPiBOOT jumper before connecting the power and USB cables to the target device.
If the device fails to connect then please see https://rpitd.co/rpiboot for debugging tips.

Waiting for BCM2835/6/7/2711/2712...

Directory not specified - trying default /usr/share/rpiboot/mass-storage-gadget64/
read_file: Failed to read "2712/bootcode5.bin" from "/usr/share/rpiboot/mass-storage-gadget64/bootfiles.bin" - No such f
ile or directory
Trying local path mass-storage-gadget64/
Sending bootcode.bin
Successful read 4 bytes
Waiting for BCM2835/6/7/2711/2712...

Second stage boot server
File read: mcb.bin
File read: memsys00.bin
File read: memsys01.bin
File read: memsys02.bin
File read: memsys03.bin
File read: bootmain
Loading: mass-storage-gadget64//config.txt
File read: config.txt
Loading: mass-storage-gadget64//boot.img
File read: boot.img
  
```

Figure 7.12. Configuring RPi CM5 as Mass Storage Device

4. The *Insert disk* window or drive content window may prompt when process is completed. Click **Cancel** or click x to close the prompt.

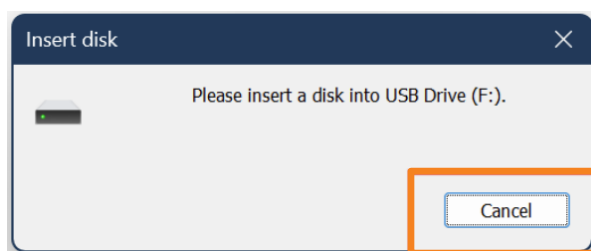


Figure 7.13. Pop Up to Insert Disk

5. Open Raspberry Pi Imager software. Ensure the **host PC** is connected to the **internet**.
6. Download the RPi OS image from RPi_OS folder listed in [Table 2.1](#). Image format is `RPi_OS-arm64_<OS_Version>_CLNX_HMI_v<Edge_Hub_Version>.img.xz`. For example: `RPi_OS-arm64_1.0.0_CLNX_HMI_v5.img.xz`.
7. Click **Choose Device** and select **Raspberry Pi 5**.
8. Click **Choose OS**. Navigate to click the **Use custom** option and select the downloaded RPi OS image.

9. Click **Choose Storage** and select **mmcblk0 USB Device – 31.3 GB**. This must appear when rpiboot process is successful.
10. Click **Next**.
11. Select **No** on the **OS Customization** settings.
12. Start flashing by clicking **Yes**.
13. The flashing process takes approximately **15 minutes** to complete.
14. This writes the OS image to the RPi CM5, followed by verification.
15. Click **Continue** after the process completes.
16. Disconnect USB Type A to Type C cable from the PC and then only remove the jumper across pin header **1 and 2** on **Jumper J10**.

References

- [CrossLinkU-NX HMI Demo](#) web page
- [CrossLinkU-NX SoM Board](#) web page
- [Lattice Insights](#) for Lattice Semiconductor training courses and learning plans
- [Lattice Radiant Programmer Tools](#) FPGA Programmer Standalone software for Windows
- [Programming Tools User Guide for Radiant Software 2025.1](#)
- [Raspberry Pi Compute Module Hardware](#)
- [Raspberry Pi Imager](#) v1.9.6 and above for Raspberry Pi's Imager tool
- [Raspberry rpiboot](#) for Raspberry Pi's rpiboot command-line utility

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Submit a technical support case through www.latticesemi.com/techsupport.

For frequently asked questions, please refer to the Lattice Answer Database at www.latticesemi.com/Support/AnswerDatabase.

Revision History

Revision 1.2, April 2026

Section	Change Summary
Introduction	Updated Features section to add bullet points in <i>Host</i> .
Functional Description	Updated Web Server section content including adding additional feature information in the section description, with enhancements such as UI improvements/graceful exits and improved wait time for inactivity from user, and updating the figure and caption of Figure 3.2. Web Server Landing Page .
sensAI Edge Vision Engine Tool	- Made some formatting updates in the following figures: Figure 4.5. Feature Enablement, Figure 4.6. Pipeline Control with Only Person Detection, Figure 4.7. Pipeline Control with Only Face Detection, Figure 4.8. Maximum IPS, Figure 4.9. Face ID on All Users Setting in Pipeline Control, Figure 4.12. Register New Face ID, and Figure 4.15. Clear Face ID Gallery. - Updated the following figures to reflect the latest codes: Figure 4.1. Landing Page Screen, Figure 4.3. Metadata Screen – Sensing Mode/ULP Sensing Mode, Figure 4.4. Metadata Presentation, Figure 4.10. Face ID on All Users, and Figure 4.11. Face ID on Main User Only. - Added the Metrics Screen section.
Running the HMI Demo Design on CrossLinkU-NX SoM HW Board	- Updated the Connecting the Raspberry Pi CM5 Network section content including adding Figure 5.19. Check Host IP Command, removing previous sub-sections, and adding RPI CM5 Terminal Access through SSH and Enabling Static IPv4 Address (Optional). - Updated Running EVE and Web Server section content including Figure 5.21. IP Address of Web Server.

Revision 1.1, February 2026

Section	Change Summary
Introduction	Updated section content to add details related to communication protocols.
Directory Structure and Files	Updated section content including Table 2.1. Folder Description.
Running the HMI Demo Design on CrossLinkU-NX SoM HW Board	- Updated Requirements section content including minor editorial fixes. - Updated Connecting the RPi CM5's Ethernet section content including updating steps and figures in sub-sections and adding additional sections in Setting Up Direct Connection to Host PC.
Debug Methodology	- Updated Figure 7.1. Debug Methodology Flow Chart. - Updated step 1 to add bullet points in Verifying Ethernet Connection. - Moved Programming the Bit File section here (previously in Running the HMI Demo Design on CrossLinkU-NX SoM HW Board). - Moved the Flashing the RPi CM5 section here (previously in Running the HMI Demo Design on CrossLinkU-NX SoM HW Board).

Revision 1.0, December 2025

Section	Change Summary
All	Initial release.



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